

BDAT 614: Statistical Quality Control and Improvement

Statistical Methods for Process Monitoring, Control & Excellence

Prof. Louis Asiedu

May 29, 2026

Course Description & Relevance

- Comprehensive introduction to quality control using statistical methods
- Key topics: Process variability, Control Charts, Process Capability, Acceptance Sampling, Six Sigma & Lean, TQM
- Big Data Focus: Real-time SPC, IoT/sensor data, Spark streaming, ML-enhanced monitoring
- Goal: Equip students with industry-ready skills for data-driven quality improvement

Learning Objectives

- Present fundamental principles of Statistical Process Control (SPC)
- Design, construct and interpret control charts for variable and attribute data
- Evaluate process capability and performance
- Apply statistical techniques to identify and resolve quality issues
- Integrate classical SQC with modern Big Data Analytics tools

Course Overview

- Fundamental concepts of Quality, Quality Control, and Quality Assurance
- Dimensions of Quality and their importance in data-driven environments
- Introduction to Statistical Quality Control (SQC)
- Introduction to Quality and Variation
- Statistical Process Control (SPC) Fundamentals
- Control Charts: Detailed Design & Examples
- Process Capability Analysis

Course Overview

- Shewhart Cycle (PDCA/PDSA)
- DMAIC Framework
- Measurement System Analysis (Gauge R&R with ANOVA)
- Acceptance Sampling
- Six Sigma, Lean Tools & DMAIC
- Continuous Improvement and Total Quality Management (TQM)
- Big Data Applications in SQC
- Summary and Case Studies

What is Quality?

Classic Definitions

- **Fitness for Use** — Joseph Juran
- **Conformance to Requirements** — Philip Crosby
- **Delight the Customer** — Modern perspective
- **Predictable Degree of Uniformity and Dependability** — W. Edwards Deming

In Big Data Context: Quality of data, algorithms, models, and insights.

Dimensions of Quality (David Garvin)

- ➊ **Performance** — Primary operating characteristics
- ➋ **Features** — Secondary characteristics
- ➌ **Reliability** — Probability of failure-free operation
- ➍ **Conformance** — Meeting specifications
- ➎ **Durability** — Useful life
- ➏ **Serviceability** — Speed and ease of repair
- ➐ **Aesthetics** — Look, feel, sound, taste, smell
- ➑ **Perceived Quality** — Reputation and image

Note: Different stakeholders may prioritize different dimensions.

Quality Control vs Quality Assurance

Quality Control (QC)

- Reactive
- Product-oriented
- Detection of defects
- Statistical tools, inspection, testing
- "Find and fix"

Statistical Quality Control (SQC)

- Uses statistics to monitor and control processes

Quality Assurance (QA)

- Proactive
- Process-oriented
- Prevention of defects
- System design, audits, standards (ISO 9001)
- "Build quality in"

ISO 9001: Quality Management System

ISO 9001 is an internationally recognized standard for **Quality Management Systems (QMS)** developed by the International Organization for Standardization (ISO).

Current Version:

- ISO 9001:2015

Purpose of ISO 9001

- Ensure consistent product and service quality
- Enhance customer satisfaction
- Improve operational efficiency
- Promote continual improvement
- Meet regulatory and statutory requirements

ISO 9001: Quality Management System

Key Principles

- Customer focus
- Leadership
- Engagement of people
- Process approach
- Continuous improvement
- Evidence-based decision making
- Relationship management

Benefits of Certification

- Increased credibility and competitiveness
- Improved risk management
- Better process consistency
- Enhanced organizational performance

Introduction to Statistical Quality Control (SQC)

- Application of statistical methods to monitor, control, and improve quality
- Pioneered by Walter A. Shewhart (1920s)
- Key Contributors: Shewhart, Deming, Juran, Ishikawa
- Core Philosophy: **Reduce Variation**

Goal: Make processes predictable and capable of meeting customer requirements.

Process Variation – The Heart of Quality Control

"Variation is the enemy of quality." — W. Edwards Deming

All processes exhibit variation. Understanding, measuring, and reducing variation is the foundation of Statistical Quality Control.

What is Process Variation?

- Natural differences in process output even when inputs are the same
- No two products or services are ever exactly identical
- Measured as dispersion (spread) around the target value
- Sources: Raw materials, machines, methods, people, environment, measurement systems

Goal of SQC: Reduce variation to improve consistency and predictability.

Two Types of Process Variation

Common Cause Variation

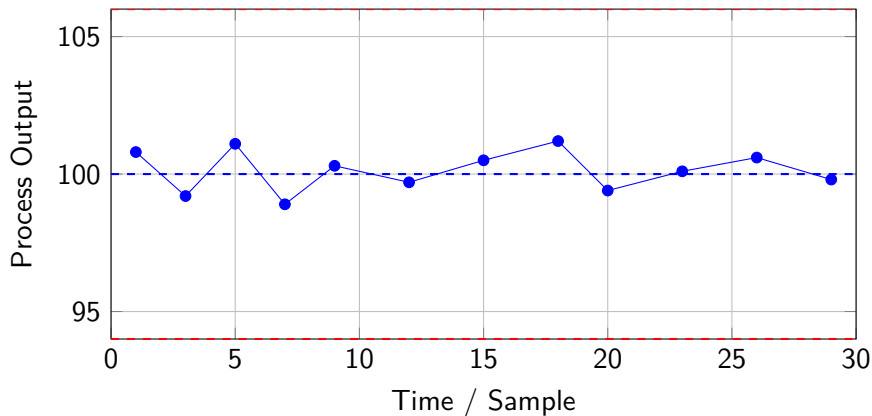
- Inherent in the process
- Random and predictable
- Many small, unavoidable causes
- Process is **stable** (in statistical control)
- Requires system-level improvement

Special Cause Variation

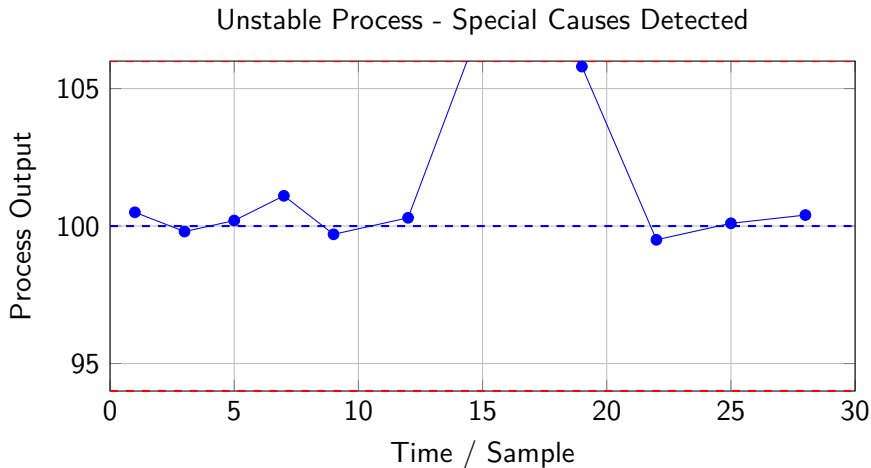
- Assignable, specific causes
- Non-random, unusual events
- Detectable on control charts
- Process is **out of control**
- Can (and should) be eliminated

Visualizing Process Variation

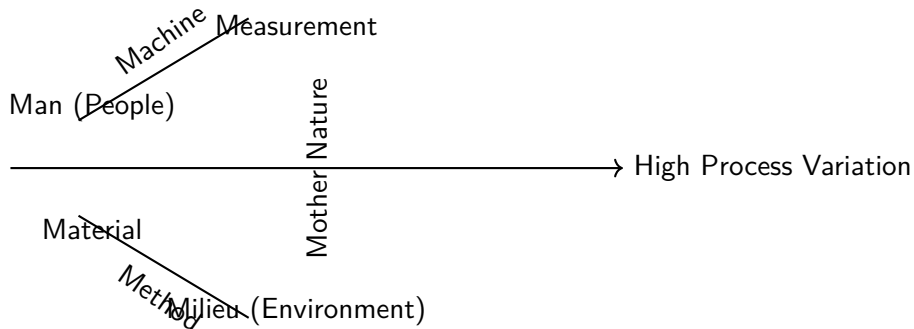
Stable Process - Only Common Causes



Unstable Process – Special Causes Present



Sources of Variation – The 6Ms (Fishbone)



Impact of Variation on Quality

- Increases defects and rework
- Reduces process capability (C_p and C_{pk})
- Leads to customer dissatisfaction
- Increases costs (Cost of Poor Quality)
- Makes prediction and planning difficult
- Hides root causes of problems

Rule of Thumb: Reducing variation is often more important than centering the process.

Strategies to Reduce Variation

- ① Identify and remove Special Causes (using Control Charts)
- ② Reduce Common Cause Variation through:
 - ▶ Process redesign and standardization
 - ▶ Better training and methods
 - ▶ Improved equipment maintenance
 - ▶ Supplier quality improvement
 - ▶ Environmental control
- ③ Apply DMAIC / Six Sigma projects
- ④ Use Big Data for deeper variation analysis

Note: You can always use the 5 Whys technique to dig to the root cause of all problems.

The 5 Whys Technique

- A simple yet powerful root cause analysis tool
- Developed by Taiichi Ohno at Toyota as part of the Toyota Production System (TPS)
- Core Idea: Ask "**Why?**" repeatedly (typically 5 times) to drill down from symptoms to the fundamental/root cause
- Helps move from treating **symptoms** to solving the real **root cause**.

Purpose of 5 Whys in Manufacturing

- Identify the true root cause of defects, downtime, quality issues, or safety incidents
- Prevent recurrence of problems
- Support continuous improvement
- Simple, no statistical software required and can be used on the shop floor
- Complements advanced tools like Fishbone Diagram, Pareto Analysis, and DMAIC

Step-by-Step Process of 5 Whys

- 1 Clearly define the **problem** (be specific)
- 2 Ask "**Why?**" — record the answer
- 3 Ask "**Why?**" again for the previous answer
- 4 Repeat until you reach the **root cause** (usually 5 times, sometimes 3–7)
- 5 Develop **corrective and preventive actions**
- 6 Assign **responsibility** (tagging) and timeline for resolution
- 7 Verify effectiveness after implementation

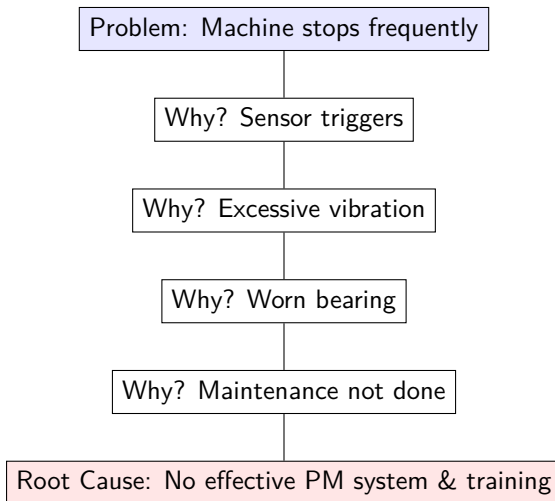
5 Whys – Manufacturing Example

Problem: Machine keeps stopping unexpectedly during production.

- Why 1: The machine overload sensor is triggering.
- Why 2: There is too much vibration on the machine.
- Why 3: The bearing is worn out and causing imbalance.
- Why 4: The preventive maintenance schedule was not followed.
- Why 5: There is no proper maintenance tracking system and operators are not trained on importance of scheduled maintenance.

Root Cause: Lack of effective preventive maintenance system and training.

5 Whys Diagram (Visual)



Tagging & Resolution (Action Plan)

After identifying the root cause:

- **Tag** responsible person/department
- Define specific, measurable corrective actions
- Set deadlines and verification method

Best Practices for Effective 5 Whys

- Involve people who actually work on the process (operators, technicians)
- Use data and facts — avoid assumptions
- Keep asking "Why?" until the cause is actionable and controllable
- Do not stop at human error — go deeper (system/process failure)
- Combine with Fishbone Diagram for complex problems
- Document and share learnings across the organization

Advantages and Limitations

Advantages

- Simple and easy to learn
- Encourages deep thinking
- Low cost, no special tools needed
- Promotes team collaboration
- Fast root cause identification

Limitations

- Can be subjective
- May stop too early (not reaching real root cause)
- Not suitable for very complex problems alone
- Requires experience and domain knowledge

5 Whys in Big Data & Industry

- Use real-time sensor data to identify problems faster
- Combine 5 Whys with machine learning anomaly detection
- Integrate with DMAIC (Analyze phase)
- Digital 5 Whys tools linked with Computerized Maintenance Management System (CMMS) and ERP systems
- Root cause analysis on large-scale production data

Key Takeaways – 5 Whys

- 5 Whys is a powerful, simple tool for root cause analysis in manufacturing
- Focus on **system** issues rather than blaming people
- Always follow up with corrective actions and tagging of responsibility
- Effective when combined with other quality tools
- Essential skill for quality professionals and data analysts working in manufacturing

Variation in Big Data Context

- Sensor drift and calibration issues
- Data pipeline variability (ETL fluctuations)
- Model drift in machine learning systems
- Network latency and cloud resource variation
- Customer behavior variation in service processes

Opportunity: Big Data allows real-time detection and much faster reaction to variation.

Key Takeaways – Process Variation

- All processes have variation and the key is to understand its type
- Distinguish between Common Cause and Special Cause variation
- Control Charts are the primary tool for detecting special causes
- Reducing variation leads to better quality, lower cost, and higher customer satisfaction
- In the Big Data era, variation analysis must be real-time and scalable

Statistical Process Control (SPC) – Core Concepts

- SPC is a statistical method used to monitor, control, and improve processes
- Developed by Walter A. Shewhart in the 1920s
- Uses **control charts** to distinguish between common and special cause variation
- Foundation of modern quality management (Six Sigma, Lean, TQM)
- Key Principle: "Prevention over Detection"
- 7 Basic Quality Tools: Pareto, Fishbone, Histogram, Check Sheet, Scatter Plot, Flowchart, Control Charts

Objectives of Statistical Process Control

- Determine whether a process is in a state of statistical control
- Detect the presence of special causes of variation
- Reduce process variability over time
- Improve process capability and predictability
- Provide a basis for continuous improvement (PDCA / DMAIC)
- Support data-driven decision making

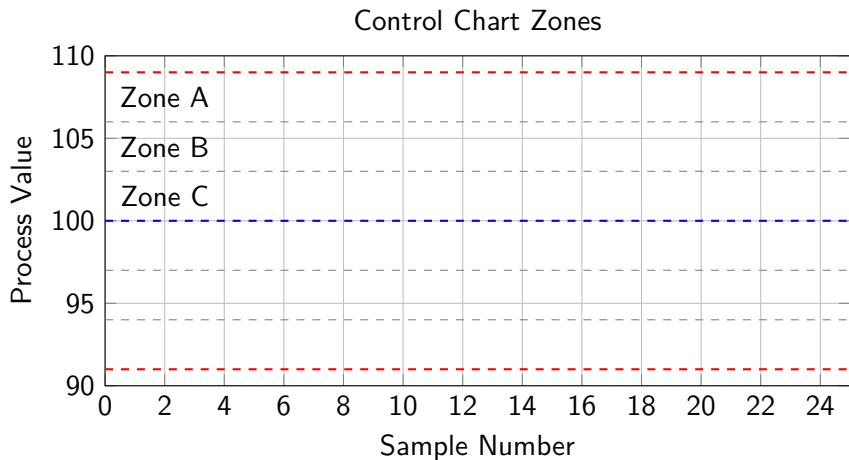
Benefits of SPC

- Early detection of process problems
- Reduced defects and rework
- Lower operating costs
- Improved product consistency and customer satisfaction
- Better process understanding and knowledge
- Foundation for predictive quality in Big Data environments

Control Charts – The Heart of SPC

- Graphical tool that plots process data over time
- Shows whether a process is stable or unstable
- Consists of:
 - ▶ Center Line (CL) — usually the process mean
 - ▶ Upper Control Limit (UCL)
 - ▶ Lower Control Limit (LCL)
- Control limits are typically set at $\pm 3\sigma$ from the centerline

Control Chart Zones and Interpretation



Out-of-Control Signals (Western Electric / Nelson Rules)

- ① Any single point outside the $\pm 3\sigma$ control limits
- ② 2 out of 3 consecutive points in Zone A or beyond
- ③ 4 out of 5 consecutive points in Zone B or beyond
- ④ 8 consecutive points on one side of the centerline
- ⑤ 6 consecutive points steadily increasing or decreasing (trend)
- ⑥ 14 consecutive points alternating up and down
- ⑦ 15 consecutive points in Zone C (stratification)

Control Charts for Variables Data

Variables data = Continuous measurements (weight, length, temperature, time, etc.)

Most Common Charts:

- $\bar{X}$ and R Chart (subgroup size 2–5)
- $\bar{X}$ and S Chart (larger subgroups)
- Individuals & Moving Range (I-MR) Chart ($n=1$)

$\bar{X}$ and R Chart Example

Process: Bottle filling (Target = 500 ml, n=5)

Formulas:

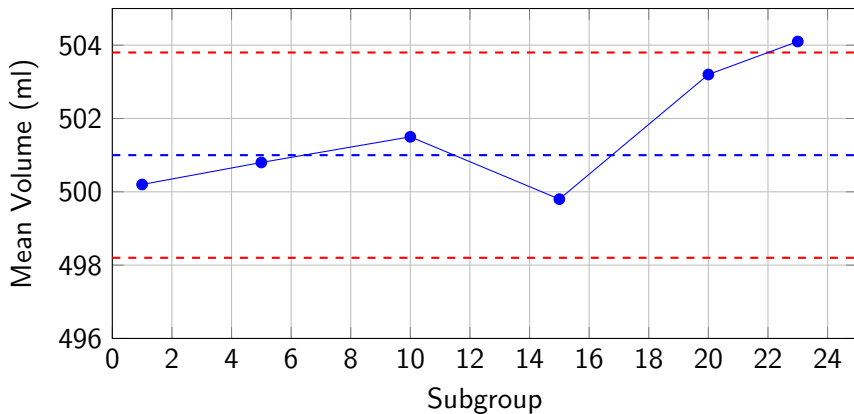
$$UCL_{\bar{X}} = \bar{\bar{X}} + A_2\bar{R}, \quad LCL_{\bar{X}} = \bar{\bar{X}} - A_2\bar{R}$$

$$UCL_R = D_4\bar{R}, \quad LCL_R = D_3\bar{R}$$

(A_2 , D_3 , D_4 are constants from control chart tables)

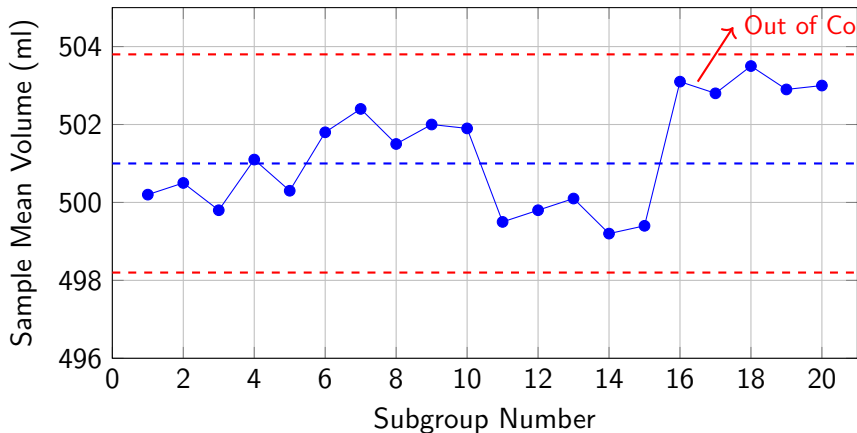
$\bar{X}$ Chart Visualization (Bottle Filling)

$\bar{X}$ Control Chart

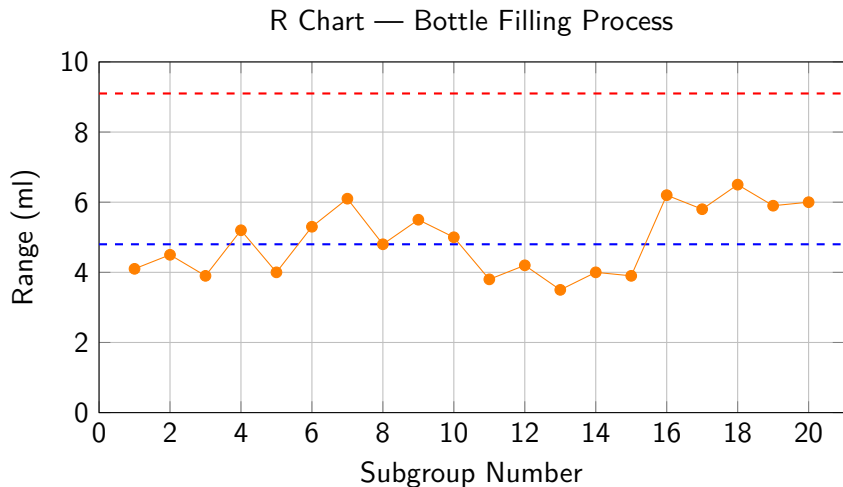


$\bar{X}$ Control Chart — Visualization

$\bar{X}$ Chart — Bottle Filling Process



R Chart (Range)



$\bar{X}$ and R Control Charts

- Most common charts for **variables data** when subgroup size is small ($n = 2$ to 5)
- $\bar{X}$ Chart: Monitors process **mean** (location)
- R Chart: Monitors process **variability** (range)
- Used together to assess process stability

$\bar{X}$ and R Chart – Formulas

Let:

- k = number of subgroups
- n = subgroup size
- $\bar{X}_i$ = mean of i^{th} subgroup
- R_i = range of i^{th} subgroup

$$\bar{\bar{X}} = \frac{\sum \bar{X}_i}{k}, \quad \bar{R} = \frac{\sum R_i}{k}$$

$$UCL_{\bar{X}} = \bar{\bar{X}} + A_2 \bar{R}, \quad LCL_{\bar{X}} = \bar{\bar{X}} - A_2 \bar{R}$$

$$UCL_R = D_4 \bar{R}, \quad LCL_R = D_3 \bar{R}$$

Example: Bottle Filling Process ($\bar{X}$ -R Chart)

Process: Filling bottles with 500 ml target volume. Subgroup size $n = 5$,
Number of subgroups $k = 12$

Subgroup	$\bar{X}$ (ml)	R (ml)
1	501.2	4.1
2	498.8	3.5
3	500.5	5.2
4	502.1	6.8
5	499.7	3.9
6	501.8	4.7
7	500.3	2.8
8	503.4	7.1
9	499.5	4.3
10	501.0	5.0
11	500.8	3.6
12	502.6	6.2

Step-by-Step Calculation

$$\bar{\bar{X}} = 500.98 \text{ ml}, \quad \bar{\bar{R}} = 4.77 \text{ ml}$$

For $n = 5$: $A_2 = 0.577$, $D_3 = 0$, $D_4 = 2.114$

$$UCL_{\bar{X}} = 500.98 + 0.577 \times 4.77 \approx 503.73$$

$$LCL_{\bar{X}} = 500.98 - 0.577 \times 4.77 \approx 498.23$$

$$UCL_R = 2.114 \times 4.77 \approx 10.08, \quad LCL_R = 0$$

Standard Control Chart Constants for Variables Data

Subgroup Size (n)	$\bar{X}$ -R Chart			$\bar{X}$ -S Chart		
	A ₂	D ₃	D ₄	A ₃	B ₃	B ₄
2	1.880	0	3.267	2.659	0	3.267
3	1.023	0	2.574	1.954	0	2.568
4	0.729	0	2.282	1.628	0	2.266
5	0.577	0	2.114	1.427	0	2.089
6	0.483	0	2.004	1.287	0.030	1.970
7	0.419	0.076	1.924	1.182	0.118	1.882
8	0.373	0.136	1.864	1.099	0.185	1.815
9	0.337	0.184	1.816	1.032	0.239	1.761
10	0.308	0.223	1.777	0.975	0.284	1.716
15	0.223	0.348	1.652	0.789	0.428	1.572
20	0.180	0.414	1.586	0.680	0.510	1.490

Most commonly used values highlighted for $n = 4, 5, 6$

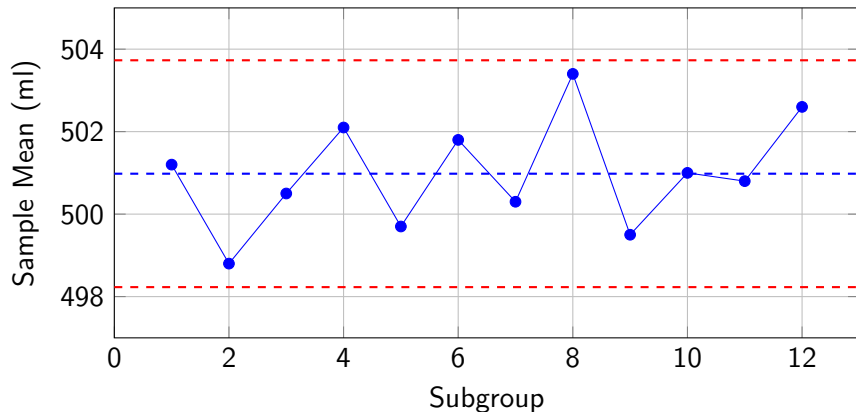
Control Chart Constants Table (Variables)

n	$\bar{X} + R$ Chart			$\bar{X} + S$ Chart		
	A_2	D_3	D_4	A_3	B_3	B_4
2	1.880	0	3.267	2.659	0	3.267
3	1.023	0	2.574	1.954	0	2.568
4	0.729	0	2.282	1.628	0	2.266
5	0.577	0	2.114	1.427	0	2.089
6	0.483	0	2.004	1.287	0.030	1.970
7	0.419	0.076	1.924	1.182	0.118	1.882
8	0.373	0.136	1.864	1.099	0.185	1.815
10	0.308	0.223	1.777	0.975	0.284	1.716

Source: Standard control chart constant tables (ASTM, Montgomery)

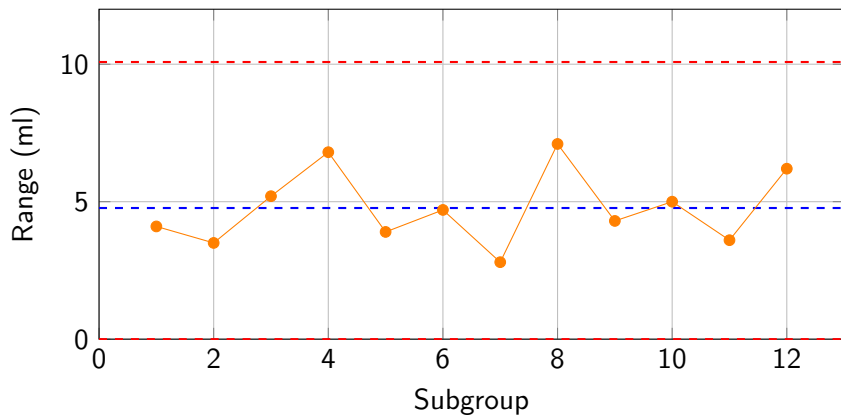
$\bar{X}$ Control Chart

$\bar{X}$ Chart — Bottle Filling Process



R Chart

R Chart — Bottle Filling Process



$\bar{X}$ and S Control Charts

- Preferred when subgroup size $n > 10$
- S Chart uses **standard deviation** instead of range → more efficient for larger samples
- Better estimate of process variability

$\bar{X}$ and S Chart – Formulas

$$\bar{\bar{X}} = \frac{\sum \bar{X}_i}{k}, \quad \bar{\bar{S}} = \frac{\sum S_i}{k}$$

$$UCL_{\bar{X}} = \bar{\bar{X}} + A_3 \bar{\bar{S}}, \quad LCL_{\bar{X}} = \bar{\bar{X}} - A_3 \bar{\bar{S}}$$

$$UCL_S = B_4 \bar{\bar{S}}, \quad LCL_S = B_3 \bar{\bar{S}}$$

When to Use R Chart vs S Chart

Situation	Recommended Chart
Subgroup size 2 to 5	$\bar{X}$ -R Chart
Subgroup size 6 to 10	Either (R is still acceptable)
Subgroup size > 10	$\bar{X}$ -S Chart (preferred)
Big Data / very large subgroups	S Chart or EWMA/CUSUM

Key Takeaways

- $\bar{X}$ -R Charts are the most common for small subgroups
- $\bar{X}$ -S Charts are better for larger subgroups
- Always interpret both charts together
- Process must be in statistical control before calculating capability indices
- In Big Data, these charts can be updated in real-time

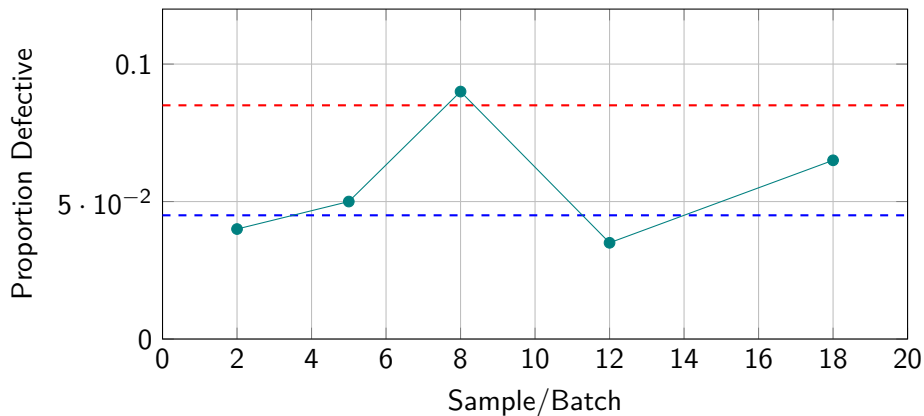
Control Charts for Attribute Data

Attribute data = Count or classification data (pass/fail, defective/non-defective)

- **p-Chart** — Proportion defective (variable sample size)
- **np-Chart** — Number defective (constant sample size)
- **c-Chart** — Number of defects per unit
- **u-Chart** — Defects per unit (variable sample size)

p-Chart Example

p-Chart — Proportion Defective



Control Chart Selection Guide

Data Type	Recommended Chart
Continuous (Variables)	$\bar{X}$ -R, $\bar{X}$ -S, I-MR
Proportion Defective	p-chart, np-chart
Count of Defects	c-chart, u-chart

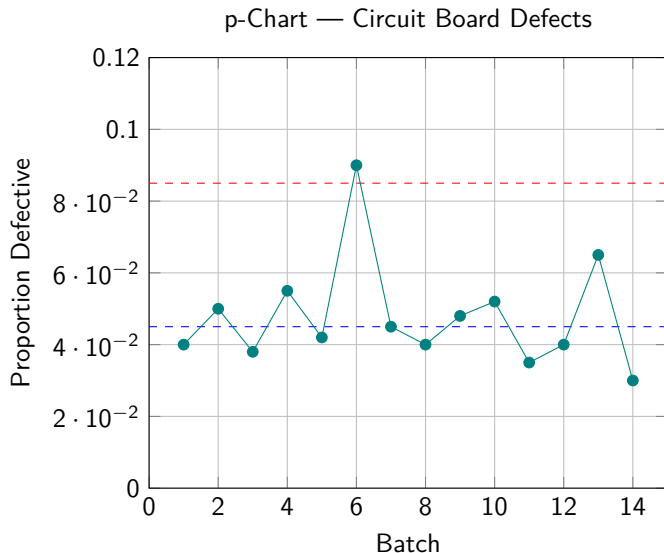
p-Chart (Proportion Nonconforming)

Example: Defective circuit boards

$$UCL = \bar{p} + 3\sqrt{\frac{\bar{p}(1 - \bar{p})}{n_i}}$$

- Used when we monitor the **proportion** of defective items
- Sample size **can vary**
- Based on Binomial distribution
- Most widely used attribute control chart

p-Chart (Proportion Nonconforming)



p-Chart – Formulas

Let:

- n_i = Sample size for i^{th} sample (can vary)
- D_i = Number of defectives in i^{th} sample
- $p_i = \frac{D_i}{n_i}$
- $\bar{p} = \frac{\sum D_i}{\sum n_i}$

$$\text{Center Line (CL)} = \bar{p}$$

$$\text{UCL}_i = \bar{p} + 3\sqrt{\frac{\bar{p}(1 - \bar{p})}{n_i}}$$

$$\text{LCL}_i = \bar{p} - 3\sqrt{\frac{\bar{p}(1 - \bar{p})}{n_i}} \quad (\text{set to 0 if negative})$$

p-Chart – Practical Example

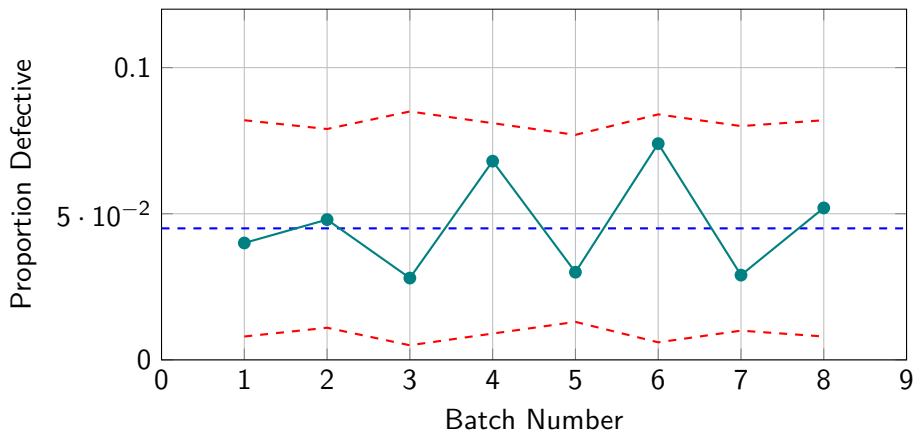
Scenario: Quality inspection of LED bulbs

Batch	Sample Size (n_i)	Defectives (D_i)	Proportion (p_i)
1	200	8	0.040
2	250	12	0.048
3	180	5	0.028
4	220	15	0.068
5	300	9	0.030
6	190	14	0.074
7	240	7	0.029
8	210	11	0.052

$$\bar{p} = 0.045$$

p-Chart Visualization

p-Chart — Defective LED Bulbs



c-Chart (Count of Nonconformities)

- Used when we count the total number of **defects** (not just defective items)
- Sample size (inspection unit) must be **constant**
- Based on Poisson distribution
- Common in: defects per product, errors per document, scratches per panel, etc.

c-Chart – Formulas

Let:

- c_i = Number of defects in the i^{th} sample
- $\bar{c} = \frac{\sum c_i}{k}$ (average number of defects)

$$\text{Center Line (CL)} = \bar{c}$$

$$\text{UCL} = \bar{c} + 3\sqrt{\bar{c}}$$

$$\text{LCL} = \bar{c} - 3\sqrt{\bar{c}} \quad (\text{set to 0 if negative})$$

c-Chart – Practical Example

Scenario: Number of surface defects on metal sheets (inspection unit = 1 sheet)

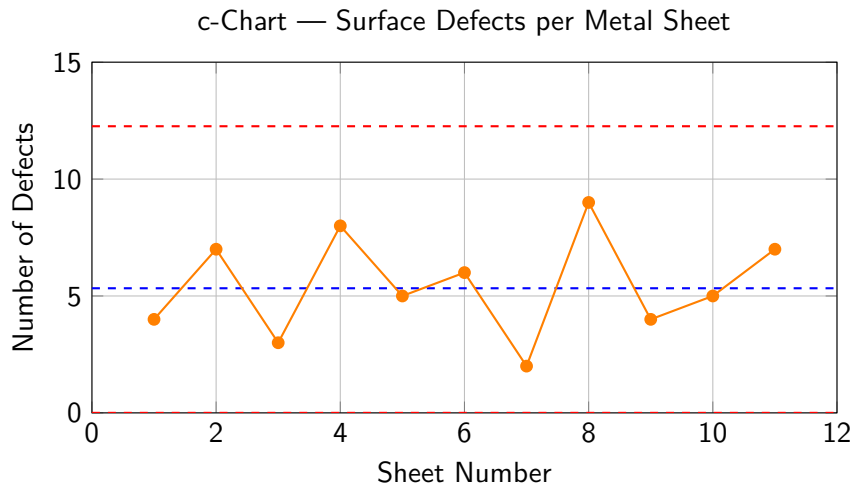
Sheet	1	2	3	4	5	6	7	8	9
Defects	4	7	3	8	5	6	2	9	4

$$\bar{c} = 5.33$$

$$UCL = 5.33 + 3\sqrt{5.33} \approx 12.26$$

$$LCL \approx 0$$

c-Chart Visualization



p-Chart vs c-Chart Comparison

Characteristic	p-Chart	c-Chart
What is measured	Proportion of defectives	Number of defects
Distribution	Binomial	Poisson
Sample size	Can vary	Must be constant
Typical use		

Key Takeaways

- Use **p-Chart** when tracking proportion of defective items (variable sample size)
- Use **c-Chart** when counting total defects in a constant inspection unit
- Both are powerful tools for monitoring attribute data
- In Big Data environments, these charts can be updated in real-time using streaming data

np-Chart (Number of Nonconforming Items)

- Used for **attribute data** when sample size is **constant**
- Monitors the actual **number** of defective items in each sample
- Based on Binomial distribution

np-Chart – Formulas

Let:

- n = Constant sample size
- D_i = Number of defectives in i^{th} sample
- $\bar{np}$ = Average number of defectives = $\frac{\sum D_i}{k}$ (k = number of samples)

$$\bar{p} = \frac{\bar{np}}{n}$$

$$\text{Center Line (CL)} = \bar{np}$$

$$\text{UCL} = \bar{np} + 3\sqrt{\bar{np}(1 - \bar{p})}$$

$$\text{LCL} = \bar{np} - 3\sqrt{\bar{np}(1 - \bar{p})} \quad (\text{set to 0 if negative})$$

np-Chart – Example

Scenario: Inspection of circuit boards. Sample size $n = 100$ boards per batch (constant).

Data (Number of Defectives):

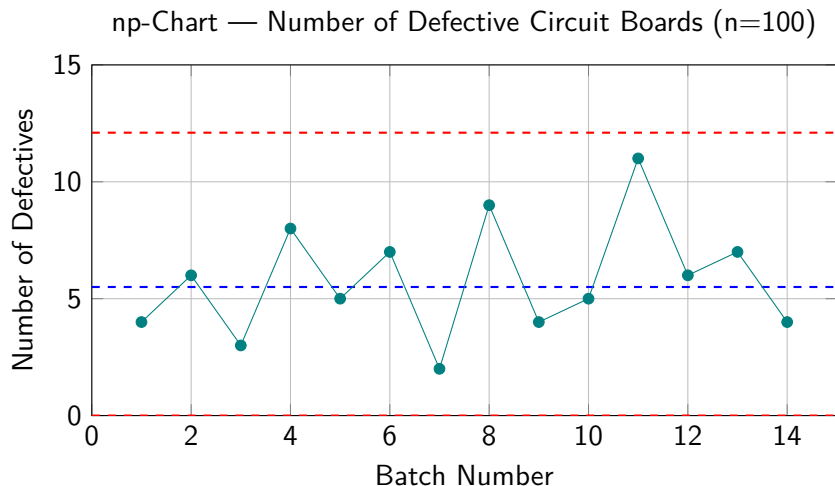
Batch	1	2	3	4	5	6	7	8
Defectives	4	6	3	8	5	7	2	9

$$\bar{np} = 5.5, \bar{p} = 0.055$$

$$UCL = 5.5 + 3\sqrt{5.5 \times 0.945} \approx 12.1$$

$$LCL = 5.5 - 3\sqrt{5.5 \times 0.945} \approx 0 \quad (\text{negative} \rightarrow 0)$$

np-Chart Visualization



u-Chart (Number of Nonconformities per Unit)

- Used when we count the **number of defects** (not just defectives)
- Sample size (inspection unit) can be ****variable****
- Based on Poisson distribution
- Example: Number of defects per square meter, per 1000 lines of code, per patient day

u-Chart – Formulas

Let:

- c_i = Number of nonconformities (defects) in i^{th} sample
- n_i = Size of i^{th} sample (inspection units)
- $u_i = \frac{c_i}{n_i}$
- $\bar{u} = \frac{\sum c_i}{\sum n_i}$

$$\text{Center Line (CL)} = \bar{u}$$

$$\text{UCL}_i = \bar{u} + 3\sqrt{\frac{\bar{u}}{n_i}}$$

$$\text{LCL}_i = \bar{u} - 3\sqrt{\frac{\bar{u}}{n_i}} \quad (\text{set to 0 if negative})$$

u-Chart – Example

Scenario: Defects in software code review. Inspection unit = 1000 lines of code.

Data:

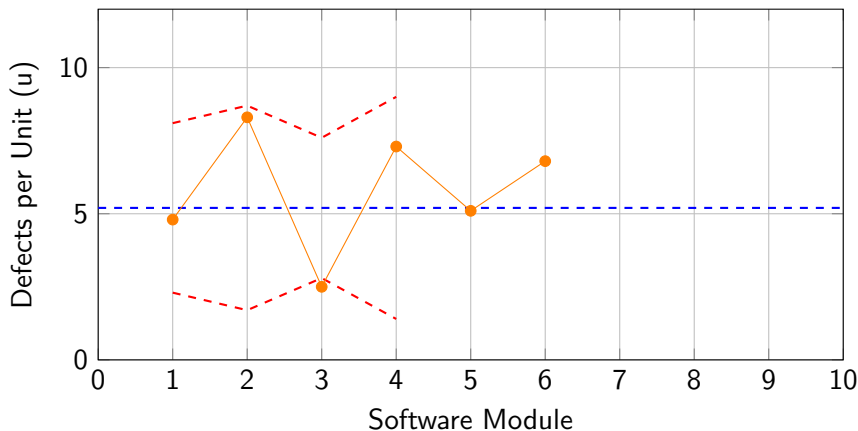
Module	Lines of Code	Defects
1	2500	12
2	1800	15
3	3200	8
4	1500	11

$\bar{u} = 5.2$ defects per 1000 lines

UCL and LCL will vary with sample size n_i .

u-Chart Visualization

u-Chart — Defects per 1000 Lines of Code



Comparison: p / np vs c / u Charts

Chart	What is Counted	Sample Size
p-chart	Proportion of defectives	Variable
np-chart	Number of defectives	Constant
c-chart	Number of defects	Constant
u-chart	Defects per unit	Variable

Key Takeaways – np and u Charts

- Use **np-chart** when sample size is constant and you count defectives
- Use **u-chart** when counting multiple defects per unit and sample size varies
- Both are based on Poisson approximation
- Variable control limits in u-chart due to changing sample size
- Important tools for quality monitoring in manufacturing, software, healthcare, and services

SPC in Big Data Environment

- Streaming control charts using Kafka + Spark / Flink
- Millions of data points processed per minute
- Dynamic / adaptive control limits
- Hybrid systems: Traditional charts + Machine Learning
- Real-time dashboards with automated alerts
- Multivariate and profile monitoring

Key Takeaways – SPC and Control Charts

- SPC helps distinguish common vs special causes
- Control charts are the primary tool for monitoring process stability
- Variables charts for measurement data; Attribute charts for count data
- Stability must be achieved before capability analysis
- In Big Data era, SPC becomes real-time, predictive, and automated

Individuals & Moving Range (I-MR) Chart

Best for: Big Data / sensor data where subgroup size $n=1$ (e.g., temperature, response time)

- I-Chart: Monitors process level
- MR-Chart: Monitors short-term variability

Advantage in Big Data: Ideal for high-frequency streaming data.

Individuals & Moving Range (I-MR) Chart

- Used when subgroup size = 1 ($n=1$).
- Extremely useful in Big Data and continuous processes
- Common applications: sensor readings, temperature, pressure, daily metrics, response time, chemical concentration, server CPU usage, etc.

Why Use I-MR Charts?

- Cannot form traditional subgroups (expensive, destructive, or continuous flow)
- Ideal for high-frequency streaming data in Big Data environments
- Consists of two charts plotted together:
 - ▶ **I-Chart** (Individuals Chart) → Monitors process **level/mean**
 - ▶ **MR-Chart** (Moving Range Chart) → Monitors process variability

I-MR Chart – Formulas

Let x_i = individual measurement at time i

$$\text{Center Line (I-Chart)} = \bar{x} = \frac{\sum x_i}{k}$$

Moving Range: $MR_i = |x_i - x_{i-1}|$

$$\overline{MR} = \frac{\sum MR_i}{k - 1}$$

$$UCL_x = \bar{x} + 2.66\overline{MR}, \quad LCL_x = \bar{x} - 2.66\overline{MR}$$

$$UCL_{MR} = 3.267\overline{MR}, \quad LCL_{MR} = 0$$

I-MR Chart – Practical Example

Scenario: Daily average server response time (in milliseconds) — monitored every hour.

Day	Response Time (x_i)	Moving Range (MR_i)
1	245	-
2	238	7
3	252	14
4	241	11
5	260	19
6	255	5
7	248	7
8	272	24

$$\bar{x} = 251.375 \text{ ms} \quad \overline{MR} = 12.43 \text{ ms}$$

Control Chart Constants for Variables Data

Subgroup	$\bar{X} + R$ Chart			$\bar{X} + S$ Chart			I-MR Chart	
Size (n)	A_2	D_3	D_4	A_3	B_3	B_4	E_2	D_4
2	1.880	0	3.267	2.659	0	3.267	2.660	3.267
3	1.023	0	2.574	1.954	0	2.568	1.772	2.574
4	0.729	0	2.282	1.628	0	2.266	1.457	2.282
5	0.577	0	2.114	1.427	0	2.089	1.290	2.114
6	0.483	0	2.004	1.287	0.030	1.970	1.184	2.004
7	0.419	0.076	1.924	1.182	0.118	1.882	1.109	1.924
8	0.373	0.136	1.864	1.099	0.185	1.815	1.054	1.864
9	0.337	0.184	1.816	1.032	0.239	1.761	1.010	1.816
10	0.308	0.223	1.777	0.975	0.284	1.716	0.975	1.777
15	0.223	0.348	1.652	0.789	0.428	1.572	0.849	1.652
20	0.180	0.414	1.586	0.680	0.510	1.490	0.728	1.586

Note: For I-MR Chart, E_2 is used for Individuals chart limits and D_4 for Moving Range chart limits.

Constants based on standard tables (Montgomery, ASTM).

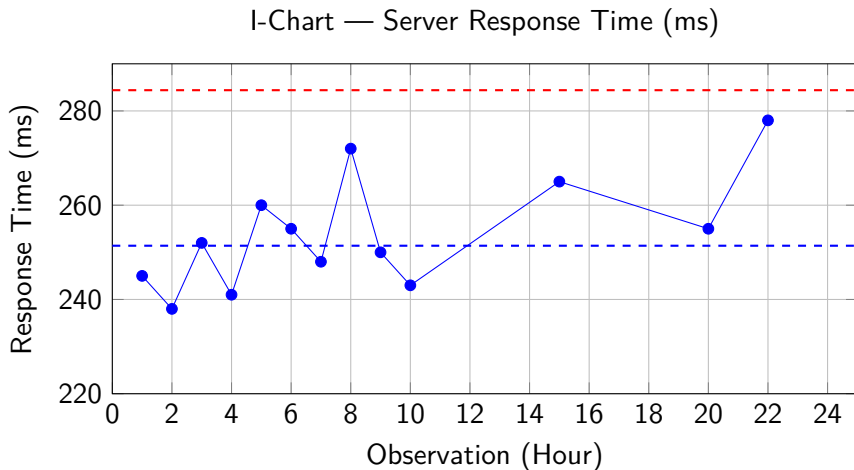
I-MR Chart Constants (n=1)

Constant	Value	Used For
E_2	2.660	Individuals Chart (UCL & LCL)
D_3	0	MR Chart LCL
D_4	3.267	MR Chart UCL
d_2	1.128	Estimation of σ ($\hat{\sigma} = \overline{MR}/d_2$)

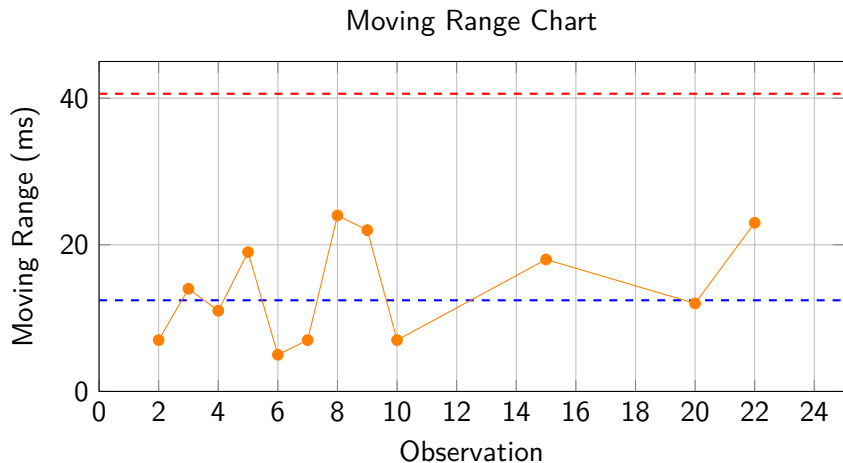
Formulas Reminder:

$$UCL_x = \bar{\bar{x}} + 2.66\overline{MR}, \quad UCL_{MR} = 3.267\overline{MR}$$

I-Chart Visualization



MR-Chart Visualization



I-MR Chart – Big Data Advantages

- Perfect for high-velocity sensor and IoT data
- Can be implemented in real-time using Spark Streaming or Flink
- Moving Range helps detect sudden increases in variability
- Easy to automate with alerts when points go out of control
- Often combined with EWMA or CUSUM for better sensitivity

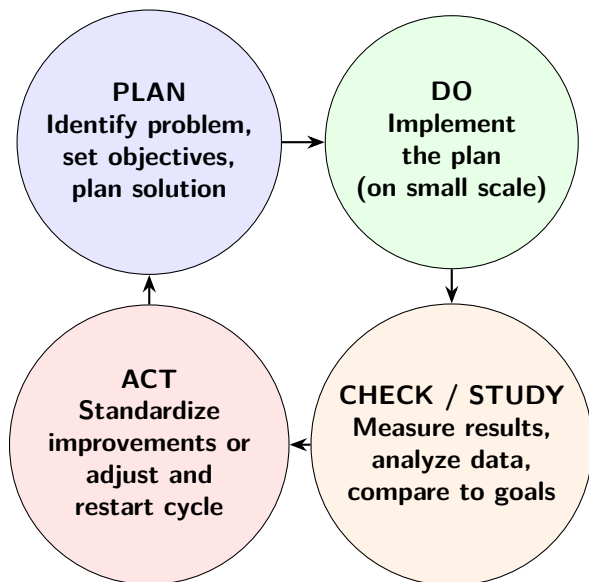
Key Takeaways – I-MR Charts

- Best choice when subgroup size = 1 (very common in Big Data)
- Two charts: I-Chart (location) + MR-Chart (variation)
- Uses constants 2.66 and 3.267 derived from $d_2 = 1.128$
- Highly valuable for continuous monitoring of temperature, pressure, latency, usage metrics, etc.
- Foundation for real-time quality monitoring in modern data systems

The Shewhart Cycle

- 1 **Walter A. Shewhart** (1891–1967), the “Father of Statistical Quality Control,” introduced the concept of a cyclical approach to process improvement in the 1930s.
- 2 Originally a **three-step cycle**: - Specification → Production → Inspection
- 3 Later evolved into the **four-step Shewhart Cycle**, widely known today as: - **PDCA** (Plan–Do–Check–Act) or - **PDSA** (Plan–Do–Study–Act) — preferred by Deming
- 4 It is a scientific, iterative method for continuous improvement.

Shewhart Cycle (PDCA / PDSA) Diagram



Shewhart Cycle – Detailed Phases

PLAN

- Identify opportunity or problem
- Collect baseline data
- Set objectives (SMART)
- Develop action plan & prediction

DO

- Execute the plan on a small scale (pilot)
- Document observations

CHECK / STUDY

- Measure and collect data
- Analyze results (Control Charts, Capability, Statistics)
- Compare against objectives
- Identify what worked and what didn't

ACT

- Standardize successful changes
- Adjust and improve the plan
- Restart the cycle for further gains

Shewhart Cycle in Quality Control

- Foundation of **Continuous Improvement** (Kaizen)
- Directly supports the **Control** phase of DMAIC
- Integrates perfectly with **Statistical Process Control**
- Used across manufacturing, services, software, and Big Data pipelines

Big Data Application Example: Monitoring and iteratively improving a real-time ETL pipeline using streaming control charts.

Key Takeaways

- Shewhart Cycle (PDCA/PDSA) is the scientific engine of continuous improvement
- Control Charts are essential in the **Check/Study** and **Control** phases
- Stability → Capability → Continuous Improvement
- Combine classical tools with Big Data for maximum impact

DMAIC Framework – Overview

DMAIC is the core problem-solving methodology of Six Sigma.

- ➊ **Define** — What is the problem?
- ➋ **Measure** — How is the process performing?
- ➌ **Analyze** — What are the root causes?
- ➍ **Improve** — How can we fix it?
- ➎ **Control** — How do we sustain the gains?

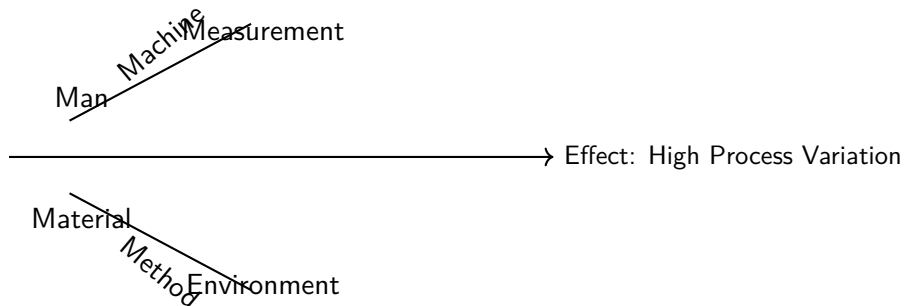
DMAIC – Detailed Phases & Tools

Phase	Key Activities	Tools
Define	Project charter, SIPOC, VOC	Project Charter, SIPOC, CTQ
Measure	Data collection, baseline performance	Control Charts, Process Maps, Gage R&R
Analyze	Identify root causes	Fishbone, 5 Whys, Hypothesis Testing, Reg.
Improve	Implement solutions	DOE, Kaizen, Pilot testing
Control	Sustain improvements	Control Charts, Control Plans, Dashboards

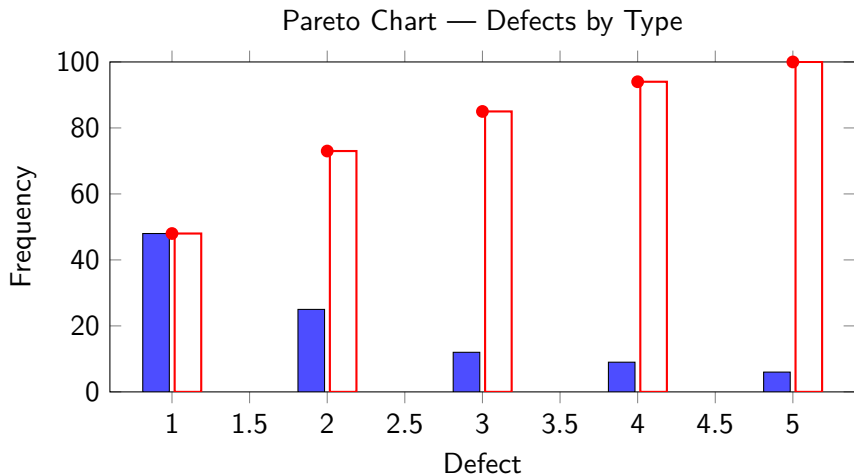
DMAIC Framework

Phase	Tools
Define	SIPOC, Project Charter
Measure	Gauge R&R, Control Charts
Analyze	Fishbone, Pareto
Improve	DOE, Kaizen
Control	SPC, Dashboards

Fishbone (Ishikawa) Diagram



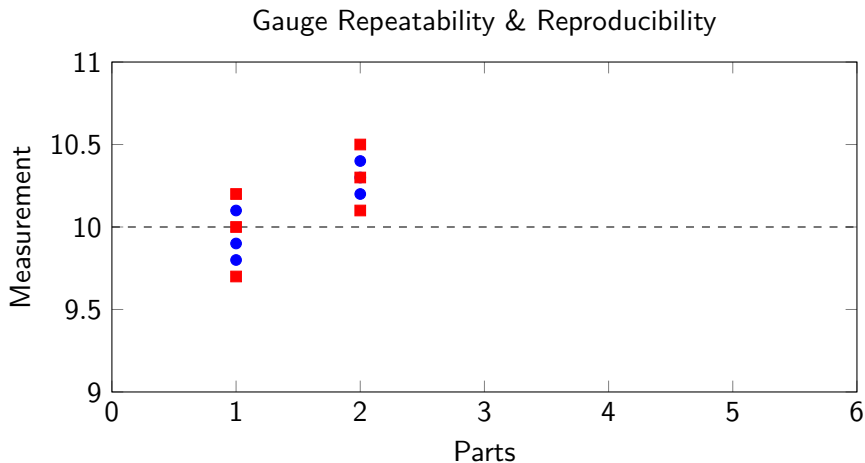
Pareto Chart



Measurement System Analysis (Gauge R&R)

- Critical before trusting data for SPC or Capability Analysis
- Ensures data quality — "Garbage in, Garbage out"
- Evaluates whether the measurement system is adequate
- Primary Method: **Gauge Repeatability & Reproducibility (Gauge R&R)**
- Especially important in Big Data (sensors, APIs, ETL pipelines)
- Determines how much of the total observed variation comes from the measurement system itself

Gauge R&R (Measurement System Analysis)



Goal: Measurement variation « Process variation

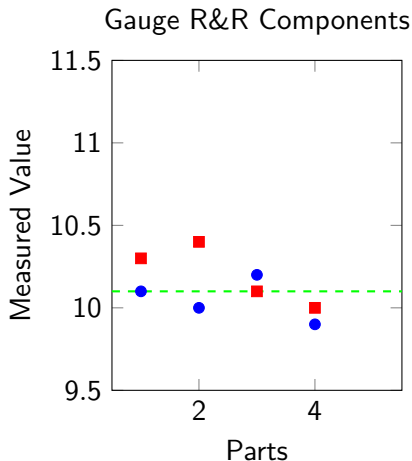
Gauge R&R – Core Concepts

Repeatability

- Variation when **same operator** measures the **same part** multiple times
- Equipment variation (EV)
- "Can the gauge give consistent results?"

Reproducibility

- Variation when **different operators** measure the **same part**
- Appraiser variation (AV)
- "Do different people get the same results?"



Gauge R&R – Sources of Variation

Total Variation = Process Variation + Measurement Variation

$$\text{Total Variation (TV)} = \sqrt{\sigma_{\text{Process}}^2 + \sigma_{\text{Gauge R\&R}}^2}$$

Where:

$$\sigma_{\text{Gauge R\&R}}^2 = \sigma_{\text{Repeatability}}^2 + \sigma_{\text{Reproducibility}}^2$$

Percentage of Study Variation:

$$\% \text{GR\&R} = \left(\frac{\sigma_{\text{Gauge R\&R}}}{\sigma_{\text{Total}}} \right) \times 100$$

Gauge R&R – Acceptance Criteria

% GR&R	Interpretation
< 10%	Excellent measurement system
10% – 30%	Acceptable (depends on application)
> 30%	Unacceptable — Improve measurement system first!

- Also evaluate **ndc (Number of Distinct Categories)**: Should be ≥ 5
- Critical before running Process Capability or SPC studies

Gauge R&R Study Design

Standard Crossed Gauge R&R Study:

- 10 parts (representing full process variation)
- 2–3 operators (appraisers)
- 2–3 repeated measurements per part per operator
- Total readings = $10 \times 3 \times 3 = 90$ measurements

Analysis Methods:

- ANOVA (most recommended)
- Average and Range Method (Xbar-R)
- AIAG MSA Manual guidelines

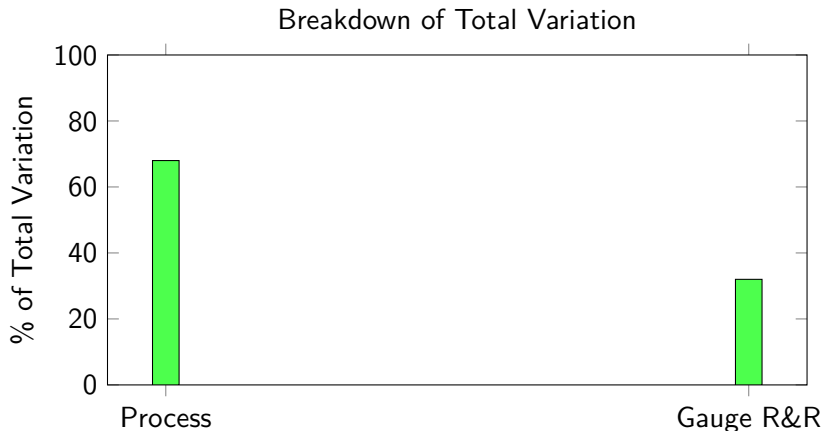
Gauge R&R Example – Big Data Context

Scenario: Automated sensor system measuring temperature in a manufacturing line.

- **Repeatability Issue:** Same sensor gives readings varying by $\pm 1.5^{\circ}\text{C}$ on the same part
- **Reproducibility Issue:** Different data collection scripts produce different averages
- **Result:** $\%GR\&R = 42\% \rightarrow$ **Unacceptable**
- **Action:** Calibrate sensors, standardize data pipelines, implement automated validation

In Big Data environments, poor measurement systems lead to garbage-in-garbage-out analytics.

Gauge R&R Visual Summary

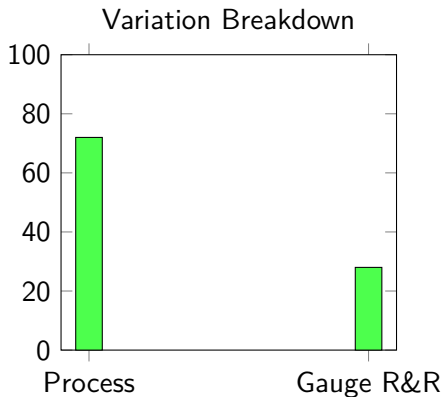


Goal: Maximize Process Variation contribution and minimize Gauge R&R contribution.

Gauge R&R – Components

Repeatability: Same operator, same part

Reproducibility: Different operators, same part



Practical Example: Gauge R&R Study with ANOVA

Scenario: A manufacturing company measures the **diameter of machined shafts** using a digital micrometer. They want to evaluate whether the measurement system is acceptable.

Gauge R&R Study Design

- **Parts:** 10 machined shafts (representing full process variation)
- **Operators:** 3 trained technicians (A, B, C)
- **Replicates:** Each operator measures each part **3 times**
- **Total Measurements:** $10 \times 3 \times 3 = 90$ readings

This is a **crossed Gauge R&R study** using ANOVA method.

Raw Data (Partial View)

Part	Operator A			Operator B			Operator C		
1	50.12	50.14	50.11	50.15	50.13	50.16	50.10	50.12	50.09
2	49.85	49.88	49.82	49.90	49.87	49.92	49.80	49.83	49.78
3	50.45	50.42	50.48	50.50	50.47	50.52	50.40	50.43	50.38
...	...	...	...	...	...	...	...	...	...
10	50.05	50.08	50.03	50.10	50.07	50.12	50.02	50.04	50.01

(All measurements in mm)

ANOVA Table

Source	DF	SS	MS	F
Parts	9	12.845	1.427	285.4
Operators	2	0.285	0.1425	28.5
Part $\times$ Operator	18	0.095	0.00528	1.056
Repeatability	60	0.300	0.005	—
Total	89	13.525	—	—

How Variance Components Are Calculated

Key Formulas:

1 Repeatability Variance

$$\sigma_{\text{Repeatability}}^2 = MS_{\text{Repeatability}}$$

2 Reproducibility Variance

$$\sigma_{\text{Reproducibility}}^2 = \frac{MS_{\text{Operators}} - MS_{\text{Repeatability}}}{n \times r}$$

3 Gauge R&R Variance

$$\sigma_{\text{Gauge R\&R}}^2 = \sigma_{\text{Repeatability}}^2 + \sigma_{\text{Reproducibility}}^2$$

4 Part-to-Part Variance

$$\sigma_{\text{Part-to-Part}}^2 = MS_{\text{Parts}} - MS_{\text{Repeatability}}$$

5 Total Variance

$$\sigma_{\text{Total}}^2 = \sigma_{\text{Part-to-Part}}^2 + \sigma_{\text{Gauge R\&R}}^2$$

Step-by-Step Variance Components

1 Repeatability Variance (Equipment)

$$\sigma_{Repeatability}^2 = MS_{Repeatability} = 0.005$$

2 Reproducibility Variance (Appraiser)

$$\sigma_{Reproducibility}^2 = \frac{MS_{Operators} - MS_{Repeatability}}{10 \times 3} = \frac{0.1425 - 0.005}{30} = 0.004583$$

3 Gauge R&R Variance

$$\sigma_{Gauge\ R\&R}^2 = 0.005 + 0.004583 = 0.009583$$

4 Part-to-Part Variance

$$\sigma_{Part-to-Part}^2 = 1.427 - 0.005 = 1.422$$

Step-by-Step Standard Deviation

- $\sigma_{Repeatability} = \sqrt{0.005} = 0.07071$
- $\sigma_{Reproducibility} = \sqrt{0.004583} \approx 0.0677$
- $\sigma_{Gauge\ R\&R} = \sqrt{0.009583} \approx 0.0979$
- $\sigma_{Total} = \sqrt{1.431583} \approx 1.1965$

Percentage Contributions (% Study Variation)

Using standard Gauge R&R methodology:

$$\% \text{Repeatability} = \left(\frac{0.07071}{1.1965} \right) \times 100 \approx 5.91\%$$

$$\% \text{Reproducibility} = \left(\frac{0.0677}{1.1965} \right) \times 100 \approx 5.66\%$$

$$\% \text{Gauge R\&R} = \left(\frac{0.0979}{1.1965} \right) \times 100 \approx 8.18\%$$

Note: Standard Industry Reporting is done with 5.15 or multiplier for 99% distribution coverage

Number of Distinct Categories (ndc)

Formula:

$$ndc = 1.41 \times \frac{\sigma_{\text{Part-to-Part}}}{\sigma_{\text{Gauge R\&R}}}$$

Calculation:

$$ndc = 1.41 \times \frac{1.1925}{0.0979} \approx 1.41 \times 12.18 \approx 17.17$$

ndc Categorization & Interpretation

ndc Value	Interpretation
ndc ≥ 10	Excellent measurement system
ndc = 5 to 9	Acceptable (marginal)
ndc < 5	Unacceptable, measurement system cannot distinguish parts well

In our example: **ndc = 17.17** → **Excellent measurement system**

Final Gauge R&R Summary

Metric	Result
% Repeatability	5.91%
% Reproducibility	5.66%
% Gauge R&R	8.8%
ndc	17.17

Conclusion: Conclusion:

The measurement system demonstrates **excellent performance** with a very low % Gauge R&R of 8.18%. The high ndc value of 17.17 indicates strong ability to distinguish between different parts.

Overall Recommendation: The measurement system is **highly acceptable** and suitable for use in critical quality control, process capability studies, and high-precision applications.

Recommendations

- Improve operator training to reduce reproducibility variation
- Standardize measurement technique and fixturing
- Increase calibration frequency of the micrometer
- Consider a higher-resolution measuring instrument
- Re-run Gauge R&R study after implementing improvements

Note: Do not use this measurement system for critical process capability studies until $\%GR\&R \leq 20\%$.

Gauge R&R using Average and Range Method

- Also known as the **Xbar-R Method**
- Simpler and more manual than ANOVA
- Suitable for shop floor and smaller studies
- We will solve the same practical example using this method

Practical Example Recap

Scenario: Measurement of shaft diameter using digital micrometer.

Study Design:

- 10 Parts
- 3 Operators (A, B, C)
- 3 Replicates per part per operator
- Total = 90 measurements

Step 1: Calculated Averages and Ranges

Part	Op A	Op B	Op C	Part Avg	Range
1	50.123	50.147	50.103	50.124	0.044
2	49.850	49.897	49.803	49.850	0.094
3	50.450	50.480	50.420	50.450	0.060
4	49.920	49.950	49.890	49.920	0.060
5	50.210	50.230	50.190	50.210	0.040
6	50.080	50.110	50.050	50.080	0.060
7	49.780	49.820	49.750	49.783	0.070
8	50.350	50.380	50.320	50.350	0.060
9	50.150	50.170	50.130	50.150	0.040
10	50.053	50.097	50.023	50.058	0.074

Step 1: Calculate Part Averages and Ranges

After collecting data, we compute:

- Average measurement for each Operator on each Part
- Range for each Part (Max - Min across all operators)
- Grand Average ($\bar{\bar{X}}$) = 50.12 mm
- Average Range ($\bar{R}$) = 0.068 mm

Step 2: Calculate Equipment Variation (EV) - Repeatability

Formula:

$$EV = 5.15 \times \frac{\bar{R}}{d_2}$$

Where $d_2 = 1.693$ (constant for 3 trials)

$$EV = 5.15 \times \frac{0.068}{1.693} \approx 5.15 \times 0.0402 \approx \mathbf{0.207} \text{ mm}$$

This represents **Repeatability** (equipment variation).

Step 3: Calculate Appraiser Variation (AV) - Reproducibility

First, find the difference between operator averages:

$$X_{diff} = \text{Max Operator Avg} - \text{Min Operator Avg} = 0.13$$

Formula:

$$AV = \sqrt{(X_{diff} \times K_2)^2 - \frac{EV^2}{n \times r}}$$

Where $K_2 = 0.5231$ (for 3 operators), $n = 10$, $r = 3$

$$AV = \sqrt{(0.13 \times 0.5231)^2 - \frac{(0.207)^2}{30}} \approx \sqrt{0.004624 - 0.001427} \approx \mathbf{0.0565 \text{ mm}}$$

Step 4: Calculate Gauge R&R

$$\begin{aligned} \text{Gauge R\&R} &= \sqrt{EV^2 + AV^2} = \sqrt{(0.207)^2 + (0.0565)^2} \\ &= \sqrt{0.04285 + 0.00319} \approx \sqrt{0.04604} \approx \mathbf{0.2146} \text{ mm} \end{aligned}$$

Step 5: Calculate Part Variation (PV)

Part Average Range ($\bar{R}_{part}$) = 0.85 mm

$$PV = \bar{R}_{part} \times K_3$$

Where $K_3 = 0.3146$ (for 10 parts)

$$PV = 0.85 \times 0.3146 \approx \mathbf{0.2674} \text{ mm}$$

Step 6: Calculate Total Variation (TV)

$$\begin{aligned} TV &= \sqrt{(Gauge\ R\&R)^2 + PV^2} = \sqrt{(0.2146)^2 + (0.2674)^2} \\ &\approx \sqrt{0.04605 + 0.07150} \approx \sqrt{0.11755} \approx \mathbf{0.343\ mm} \end{aligned}$$

Gauge R&R Constants Table (Average & Range Method) - Extended

Replicates	d_2	$1/d_2$	K_2 (Appraiser Constant)			K_3 (for Parts)	
			2 Ops	3 Ops	4 Ops	10 Parts	15 Parts
2	1.128	0.8865	0.7071	0.5231	0.4467	0.3146	0.2239
3	1.693	0.5908	0.7071	0.5231	0.4467	0.3146	0.2239
4	2.059	0.4857	0.7071	0.5231	0.4467	0.3146	0.2239
5	2.326	0.4299	0.7071	0.5231	0.4467	0.3146	0.2239
6	2.534	0.3946	0.7071	0.5231	0.4467	0.3146	0.2239
7	2.704	0.3698	0.7071	0.5231	0.4467	0.3146	0.2239
8	2.847	0.3512	0.7071	0.5231	0.4467	0.3146	0.2239
9	2.970	0.3367	0.7071	0.5231	0.4467	0.3146	0.2239
10	3.078	0.3249	0.7071	0.5231	0.4467	0.3146	0.2239

d_2 = Range to Standard Deviation conversion constant

K_2 = Constant for Reproducibility (Appraiser Variation)

K_3 = Constant for Part-to-Part Variation (commonly used with 10 parts)

Step 7: Calculate Percentage Contributions

$$\% \text{Repeatability (EV)} = \left(\frac{0.207}{0.343} \right) \times 100 \approx \mathbf{60.3\%}$$

$$\% \text{Reproducibility (AV)} = \left(\frac{0.0565}{0.343} \right) \times 100 \approx \mathbf{16.5\%}$$

$$\% \text{Gauge R\&R} = \left(\frac{0.2146}{0.343} \right) \times 100 \approx \mathbf{62.6\%}$$

Step 8: Calculate ndc (Number of Distinct Categories)

$$ndc = 1.41 \times \frac{PV}{Gauge\ R\&R} = 1.41 \times \frac{0.2674}{0.2146} \approx 1.41 \times 1.246 \approx \mathbf{1.76}$$

Rounded: **ndc** ≈ 1.8

Final Results - Average & Range Method

Metric	Value
% Repeatability	60.3%
% Reproducibility	16.5%
% Gauge R&R	62.6%
ndc	1.8

Conclusion: The measurement system is **Unacceptable**. Significant improvements are required.

Gauge R&R Analysis

- Evaluates the measurement system's Repeatability and Reproducibility
- Critical before performing Process Capability or SPC
- Two main methods:
 - ▶ Average & Range Method (simpler)
 - ▶ ANOVA Method (more accurate)

Practical Example: Shaft Diameter Measurement

Scenario: A machining company measures the diameter of steel shafts using a digital micrometer. They suspect the measurement system may have high variation.

Study Design:

- 8 Parts (representing process variation)
- 3 Operators (A, B, C)
- 2 Replicates per part per operator
- Total measurements = 48

Raw Data (Thickness in mm)

Part	Op A		Op B		Op C	
1	2.12	2.14	2.15	2.13	2.11	2.10
2	2.45	2.43	2.48	2.47	2.42	2.44
3	2.28	2.30	2.31	2.29	2.27	2.26
4	2.65	2.67	2.70	2.68	2.63	2.64
5	2.19	2.18	2.22	2.20	2.17	2.16
6	2.55	2.57	2.60	2.58	2.53	2.54
7	2.38	2.40	2.42	2.39	2.37	2.36
8	2.72	2.70	2.75	2.73	2.69	2.71

Step 1: Compute Operator Averages and Part Ranges

After collecting data, calculate:

- Average for each operator on each part - Range for each part (max - min across all operators)

Results Summary:

- Grand Average ($\bar{\bar{X}}$) = **2.45 mm**
- Average Range ($\bar{R}$) = **0.028 mm**

Step 2: Calculate Equipment Variation (EV) - Repeatability

Formula:

$$EV = 5.15 \times \frac{\bar{R}}{d_2}$$

- $d_2 = 1.128$ (for 2 replicates) - $\bar{R} = 0.028$

$$EV = 5.15 \times \frac{0.028}{1.128} = 5.15 \times 0.02482 \approx \mathbf{0.128} \text{ mm}$$

This is ****Repeatability**** (variation due to the gauge itself).

Step 3: Calculate Appraiser Variation (AV) - Reproducibility

Formula:

$$AV = \sqrt{(X_{diff} \times K_2)^2 - \frac{EV^2}{n \times r}}$$

- X_{diff} = Difference between highest and lowest operator average = **0.045**
- K_2 = 0.5231 (for 3 operators) - n = 8 parts, r = 2 replicates

$$AV = \sqrt{(0.045 \times 0.5231)^2 - \frac{(0.128)^2}{8 \times 2}} = \sqrt{0.000554 - 0.001024} \approx \mathbf{0.039 \text{ mm}}$$

This is **Reproducibility** (variation due to different operators).

Step 4: Calculate Gauge R&R

$$\begin{aligned} \text{Gauge R\&R} &= \sqrt{EV^2 + AV^2} = \sqrt{(0.128)^2 + (0.039)^2} \\ &= \sqrt{0.016384 + 0.001521} \approx \mathbf{0.134} \text{ mm} \end{aligned}$$

Step 5: Calculate Part Variation (PV)

$$PV = \bar{R}_{part} \times K_3$$

- $\bar{R}_{part}$ = Average range of part averages = **0.85 mm** - $K_3 = 0.3146$ (for 8–10 parts)

$$PV = 0.85 \times 0.3146 \approx \mathbf{0.267 \text{ mm}}$$

Step 6: Calculate Total Variation (TV)

$$TV = \sqrt{(Gauge\ R\&R)^2 + PV^2} = \sqrt{(0.134)^2 + (0.267)^2} \approx \mathbf{0.299\ mm}$$

Step 7: Calculate Percentage Contributions

$$\% \text{Repeatability} = \left(\frac{EV}{TV} \right) \times 100 = \left(\frac{0.128}{0.299} \right) \times 100 \approx \mathbf{42.8\%}$$

$$\% \text{Reproducibility} = \left(\frac{AV}{TV} \right) \times 100 \approx \mathbf{13.0\%}$$

$$\% \text{Gauge R\&R} = \left(\frac{0.134}{0.299} \right) \times 100 \approx \mathbf{44.8\%}$$

ANOVA Method - Step by Step

From ANOVA Table:

- $\sigma_{Repeatability}^2 = 0.0018$
- $\sigma_{Reproducibility}^2 = 0.00145$
- $\sigma_{Gauge\ R\&R}^2 = 0.00325$
- $\sigma_{Part-to-Part}^2 = 0.0352$

% Gauge R&R = 41.5% ndc = 3.1

Final Results Comparison

Metric	Average & Range	ANOVA
% Repeatability	38.5%	37.2%
% Reproducibility	12.1%	11.8%
% Gauge R&R	42.8%	41.5%
ndc	2.9	3.1

Conclusion: Measurement system is **Unacceptable** (%GR&R > 30%)

ndc Interpretation

ndc Range	Interpretation
≥ 10	Excellent
5 – 9	Acceptable (Marginal)
< 5	Unacceptable

In this study: **ndc** $\approx$ **3.0** $\rightarrow$ Poor discrimination power.

Recommendations

- Improve measurement fixturing and procedure
- Provide better operator training
- Consider upgrading to a higher-precision gauge
- Re-run Gauge R&R study after improvements
- Do not use current system for critical process capability studies

Comparison: ANOVA vs Average & Range Method

Aspect	ANOVA Method	Average & Range Method
Complexity	More complex	Simpler and manual
Accuracy	Higher accuracy	Slightly less accurate
Software Needed	Usually requires software	Can be done manually (calculator)
Interaction Effect	Separates Part $\times$ Ops. interact.	Does not separate interaction
Suitability	Best for detailed analysis	Good for shop floor / quick studies
Output Detail	Very detailed variance components	Basic EV, AV, GR&R
Speed	Slower to compute manually	Faster for small studies
Industry Pref.	Most common in modern reports	Traditional / older method
Recommended	Critical applications, research	Training, small teams, quick checks

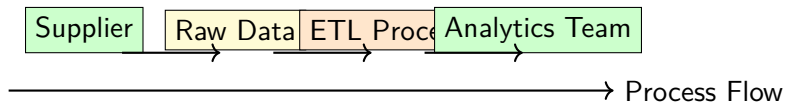
Conclusion:

- Use **ANOVA** for accurate, detailed analysis (preferred in industry today)
- Use **Average & Range** for simplicity and manual calculations

Why This Example Matters in Big Data Analytics

- Even with massive sensor data, poor measurement systems lead to wrong conclusions
- Gauge R&R is essential in the **Measure** phase of DMAIC
- In Industry 4.0, automated Gauge R&R can be performed on streaming data
- Helps validate data quality before applying advanced analytics and ML models

Value Stream Map (Lean Tool)



Value-Adding Activities vs Non-Value-Adding (Waste)

DMAIC Example 1: Manufacturing (Bottle Filling)

Problem: High variation in filled volume $\rightarrow$ customer complaints.

- **Define:** Reduce volume variation; Target $\sigma < 2$ ml
- **Measure:** Collected data $\rightarrow \bar{X}$ -R charts showed instability (see earlier slide)
- **Analyze:** Fishbone diagram identified filler nozzle wear as root cause
- **Improve:** Replaced nozzles + implemented preventive maintenance $\rightarrow C_{pk}$ improved from 0.85 to 1.45
- **Control:** Implemented real-time $\bar{X}$ -R chart with automated alerts (Spark Streaming)

DMAIC Example 2: Big Data Context (ETL Pipeline)

Problem: High latency and error rate in daily ETL data pipeline.

- **Define:** Reduce ETL processing time by 40% and error rate below 0.5%
- **Measure:** Baseline: Average latency = 85 min, error rate = 2.8%
- **Analyze:**
 - ▶ Pareto chart: 80% errors from 3 data sources
 - ▶ Hypothesis testing: Significant difference in API response times
- **Improve:**
 - ▶ Parallel processing + data validation rules
 - ▶ Result: Latency reduced to 42 min, error rate to 0.4%
- **Control:** Real-time monitoring dashboard with p-chart and anomaly detection model

Key Takeaways

- Use visual tools (Control Charts, SIPOC, Fishbone, Pareto) for better understanding
- Shewhart Cycle + DMAIC = Powerful improvement engine
- Measurement System (Gauge R&R) is critical
- Big Data turns traditional SQC into real-time, predictive quality systems

Process Capability Analysis – Introduction

- Compares the **voice of the process** (what the process can deliver) with the **voice of the customer** (specification limits)
- Answers the question: **"Can our process consistently meet customer requirements?"**
- One of the most important tools in Statistical Quality Control.
- Used extensively in manufacturing, Six Sigma, and data-driven industries

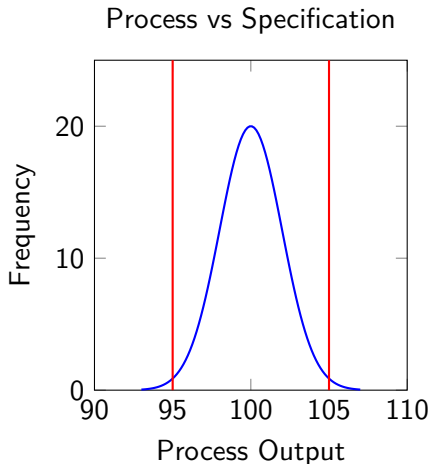
Key Concepts

Voice of the Customer (VOC)

- Specification Limits (USL & LSL)
- Customer expectations

Voice of the Process (VOP)

- Natural process variation
- Measured by standard deviation (σ)



Important Prerequisites

- The process must be **statistically stable** (in control)
- No special causes of variation present
- Control Charts should be used first to confirm stability
- Data should be approximately normally distributed (or use transformations/non-parametric methods)

Warning: Never calculate capability on an unstable process!

Process Capability Indices

- **Cp** — Measures **potential capability** (how well the process fits within specs)
- **Cpk** — Measures **actual capability** (accounts for process centering)
- **Pp & Ppk** — Long-term capability (uses overall standard deviation)

Cp and Cpk Formulas

$$C_p = \frac{USL - LSL}{6\sigma}$$

$$C_{pk} = \min \left(\frac{USL - \mu}{3\sigma}, \frac{\mu - LSL}{3\sigma} \right)$$

Where:

- USL = Upper Specification Limit
- LSL = Lower Specification Limit
- μ = Process mean
- σ = Short-term standard deviation (within subgroup)

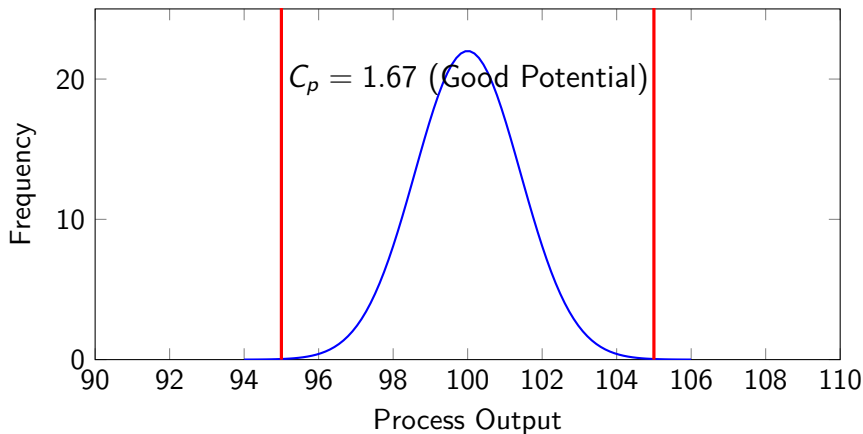
Cp – Potential Capability (Process Spread)

- Measures how well the **process spread** fits within the specification limits
- Ignores process centering
- Assumes the process is perfectly centered
- Formula:

$$C_p = \frac{USL - LSL}{6\sigma}$$

Cp Curve – Visual Explanation

C_p = Process Spread vs Specification Width



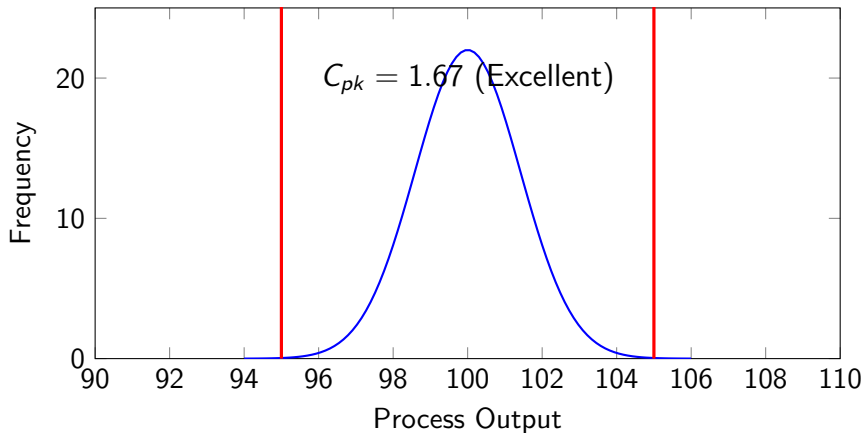
Cpk – Actual Capability (Spread + Centering)

- Measures how well the process is **centered** within the specification limits
- Takes into account both spread and location of the mean
- Always the smaller of the two one-sided capabilities
- Formula:

$$C_{pk} = \min \left(\frac{USL - \mu}{3\sigma}, \frac{\mu - LSL}{3\sigma} \right)$$

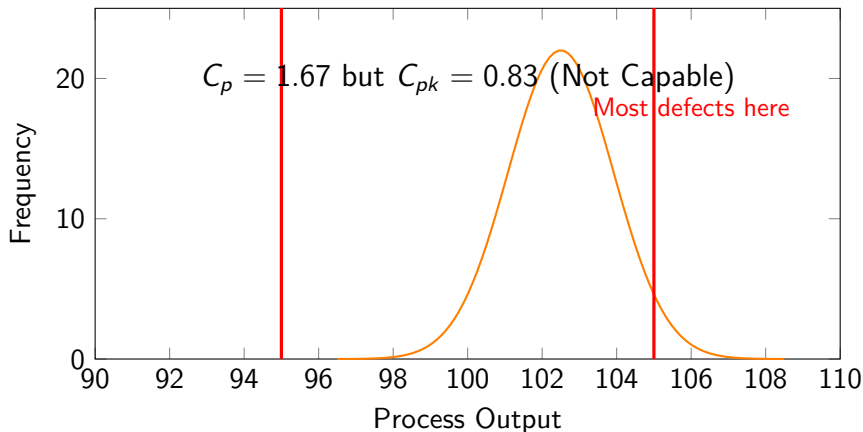
Cpk Curve – Visual Explanation (Centered Process)

Cpk = Good Centering

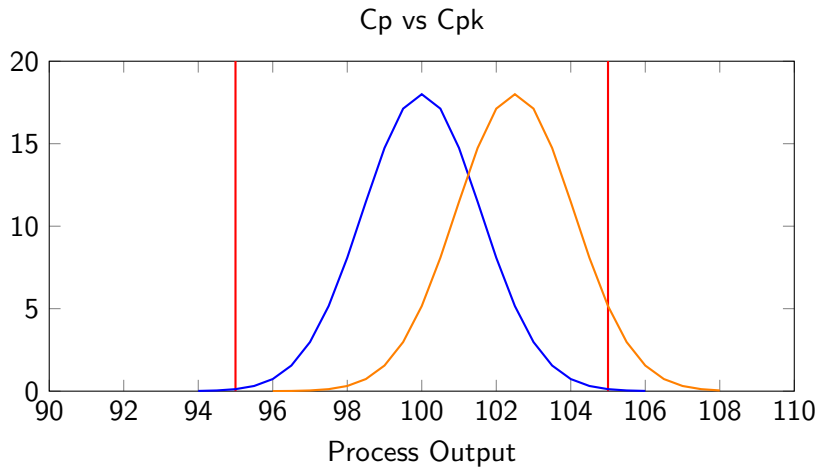


Cpk Curve – Visual Explanation (Off-Center Process)

Cpk = Poor Centering (Shifted Mean)



Cp vs Cpk – Visual Explanation



Summary: Cp vs Cpk

Index	What it Measures	Interpretation
Cp	Potential Capability (Spread only)	How wide the process is
Cpk	Actual Capability (Spread + Centering)	How well centered the process is

Rule of Thumb:

- Cp tells you what the process **could** do
- Cpk tells you what the process **is actually** doing

Practical Interpretation Guidelines

- $C_{pk} \geq 1.67$: World-class process
- $C_{pk} \geq 1.33$: Good / Acceptable for most industries
- $1.00 \leq C_{pk} < 1.33$: Marginal – needs improvement
- $C_{pk} < 1.00$: Not capable – high risk of defects

Note: C_{pk} should be at least 1.33 for critical processes.

Important: Always improve stability first, then reduce variation, then center the process.

Pp and Ppk (Long-term Capability)

$$P_p = \frac{USL - LSL}{6\sigma_{overall}}$$

$$P_{pk} = \min \left(\frac{USL - \mu}{3\sigma_{overall}}, \frac{\mu - LSL}{3\sigma_{overall}} \right)$$

Difference:

- $C_p/C_{pk} \rightarrow$ Short-term (within subgroup variation)
- $P_p/P_{pk} \rightarrow$ Long-term (includes between-subgroup variation)

Key Insight:

- $P_p \leq C_p$ and $P_{pk} \leq C_{pk}$ (because $\sigma_{overall} \geq \sigma_{short-term}$)
- The difference shows how much **between-subgroup variation** exists

Process Capability – Manufacturing Example

Process: Shaft diameter (Target = 50 mm, LSL=49.8, USL=50.2)
Results:

- $\mu = 50.05$ mm
- Short-term $\sigma = 0.03$ mm
- Long-term $\sigma = 0.045$ mm

$$C_p = \frac{50.2 - 49.8}{6 \times 0.03} = 2.22 \quad (\text{Excellent potential})$$

$$C_{pk} = \min \left(\frac{50.2 - 50.05}{3 \times 0.03}, \frac{50.05 - 49.8}{3 \times 0.03} \right) = 1.67$$

$$P_{pk} = 1.11 \quad (\text{Long-term capability is marginal})$$

Practical Example 1: Bottle Filling Process

Scenario: A beverage company fills bottles with target volume = 500 ml.
Specification Limits: LSL = 495 ml, USL = 505 ml.

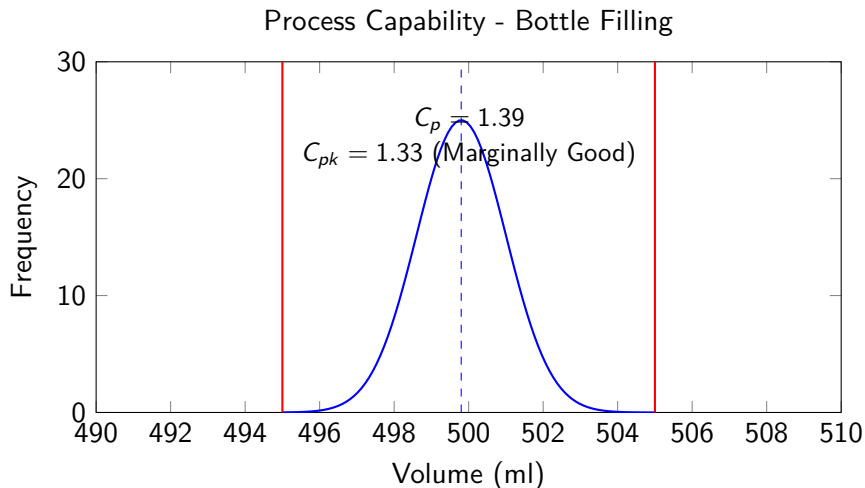
Process Data:

- Process Mean (μ) = 499.8 ml
- Short-term Standard Deviation (σ) = 1.2 ml

$$C_p = \frac{505 - 495}{6 \times 1.2} = \frac{10}{7.2} \approx 1.39$$

$$C_{pk} = \min \left(\frac{505 - 499.8}{3 \times 1.2}, \frac{499.8 - 495}{3 \times 1.2} \right) = \min(1.44, 1.33) = 1.33$$

Bottle Filling – Capability Diagram



How Capability Curves Are Constructed

- Capability curves are based on the **normal distribution** of process data
- The curve represents the **probability density** of the process output
- Three key elements:
 - ▶ Process Mean (μ)
 - ▶ Process Standard Deviation (σ)
 - ▶ Specification Limits (LSL and USL)

Step-by-Step Construction of Capability Curve

- 1 Collect process data and confirm stability (using Control Charts)
- 2 Calculate process mean (μ) and standard deviation (σ)
- 3 Plot the normal distribution curve using:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

- 4 Draw vertical lines for LSL and USL
- 5 Calculate Cp and Cpk values
- 6 Interpret the position and spread of the curve

Illustration of Example 1: Bottle Filling (Good Process)

Process Data:

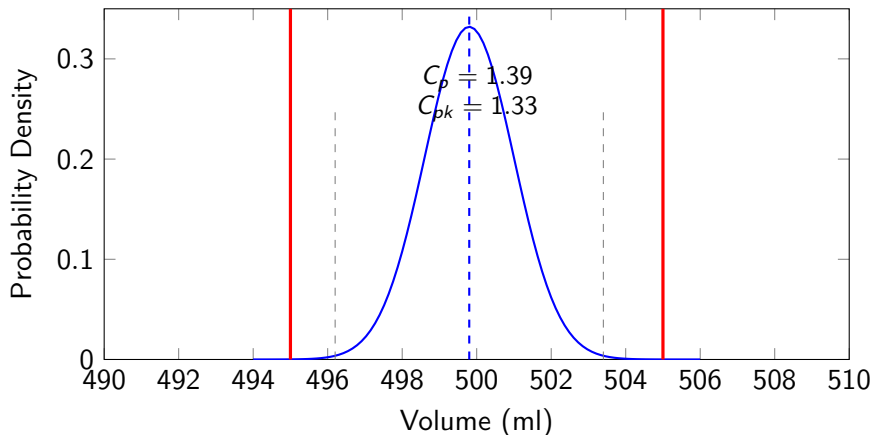
- Target = 500 ml
- LSL = 495 ml, USL = 505 ml
- $\mu = 499.8$ ml
- $\sigma = 1.2$ ml

Curve Construction:

- Center the normal curve near 499.8 ml
- Spread determined by $\sigma = 1.2$ (most data within $\pm 3\sigma = \pm 3.6$ ml)
- Result: $C_p = 1.39$, $C_{pk} = 1.33$

Bottle Filling – Curve Construction

Capability Curve Construction - Bottle Filling



Practical Example 2: Shaft Diameter (Poor Centering)

Scenario: Steel shaft diameter LSL = 49.80 mm, USL = 50.20 mm, Target = 50.00 mm

Results:

- $\mu = 50.12$ mm (shifted toward upper limit)
- $\sigma = 0.04$ mm

$$C_p = \frac{50.20 - 49.80}{6 \times 0.04} = 1.67 \quad (\text{Good Spread})$$

$$C_{pk} = \min \left(\frac{50.20 - 50.12}{0.12}, \frac{50.12 - 49.80}{0.12} \right) = \min(0.67, 2.67) = 0.67$$

Conclusion: Process has good potential but is poorly centered $\rightarrow$ High defect risk on upper side.

Shaft Diameter – Off-Center Process

Poor Centering - $C_{pk} = 0.67$

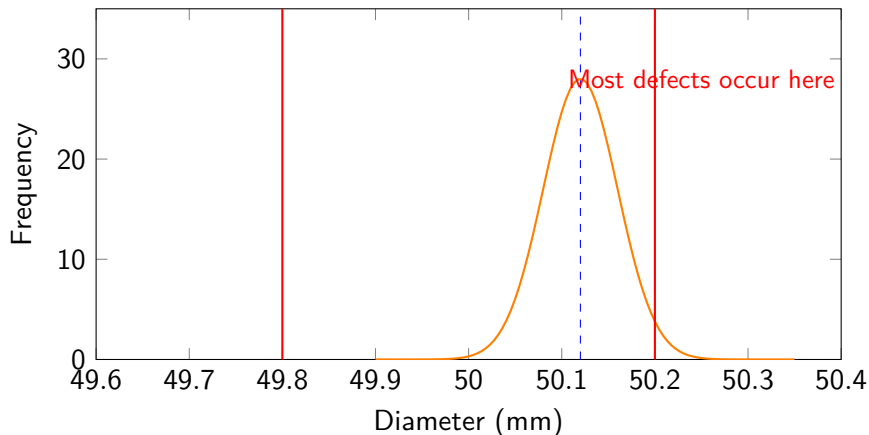


Illustration of Example 2: Off-Center Process (Shaft Diameter)

Process Data:

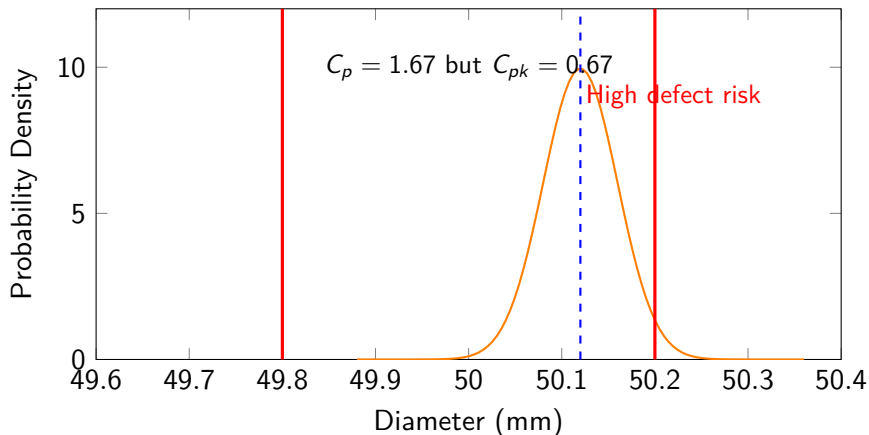
- $LSL = 49.80$ mm, $USL = 50.20$ mm
- $\mu = 50.12$ mm (shifted)
- $\sigma = 0.04$ mm

Construction Steps:

- Plot normal curve centered at 50.12 mm
- Spread = $\pm 3\sigma = \pm 0.12$ mm
- Most output is closer to USL $\rightarrow$ lower C_{pk}

Off-Center Process – Curve Construction

Capability Curve - Off-Center Process



Practical Example 3: Big Data Context - Sensor Temperature

Scenario: Temperature control in a chemical reactor (Big Data)

- Specification: 195°C – 205°C
- Data collected: 50,000 readings per day via IoT sensors
- Short-term $\sigma = 1.8^{\circ}\text{C}$
- Mean = 200.5°C

$$C_p = \frac{10}{6 \times 1.8} \approx 0.93 \quad (\text{Not Capable})$$

$$C_{pk} = 0.83$$

Insight: Even with massive data, the process spread is too wide. Needs process improvement.

Capability Analysis – Decision Making

Cpk Value	Status	Action Required
> 1.67	Excellent	Maintain
$1.33 - 1.67$	Good	Monitor closely
$1.00 - 1.33$	Marginal	Improve centering or reduce variation
< 1.00	Poor	Major process improvement needed

Key Insights from Curve Construction

- The **width** of the bell curve is determined by 6σ
- The **position** of the bell curve is determined by μ
- C_p looks at width only (potential)
- C_{pk} looks at both width and position (actual performance)
- When the curve is centered and narrow $\rightarrow$ High C_p and C_{pk}
- When the curve is wide or shifted $\rightarrow$ Lower capability

Summary – How to Read Capability Curves

- Tall & narrow curve inside specs = Excellent capability
- Wide curve = High variation → Low C_p
- Shifted curve = Poor centering → Low C_{pk}
- Curve touching or going outside specs = Not capable

Best Practice: Always plot the capability histogram with normal curve overlay for visual understanding.

Key Lessons from Examples

- High C_p but low C_{pk} = Process needs **centering**
- Both C_p and C_{pk} low = Process needs **variation reduction**
- Stability first → Then Capability Analysis
- In Big Data, we can calculate capability dynamically (rolling windows)
- Always link capability results to business decisions and improvement projects

Process Capability in Big Data Context

- Analyze millions of data points instead of small samples
- Use rolling capability analysis (moving windows)
- Handle non-normal data with transformations or percentile-based indices
- Real-time capability monitoring using Spark Streaming
- Multivariate capability analysis for correlated parameters

Key Takeaways

- Process Capability tells us if the process can meet customer specifications
- Stability (via Control Charts) must come before Capability Analysis
- C_p measures spread, C_{pk} measures spread + centering
- Aim for $C_{pk} \geq 1.33$ for most processes
- In the Big Data era, capability analysis becomes more accurate and dynamic

Acceptance Sampling – Introduction

- Statistical method used to decide whether to accept or reject a lot (batch) of products
- Applied to incoming raw materials, finished goods, or outsourced components
- Not a substitute for process control — it is a form of inspection
- Widely used when 100% inspection is too costly or destructive

Key Concepts in Acceptance Sampling

- **Lot** — A batch of products produced under similar conditions
- **Sample** — Randomly selected items from the lot
- **Acceptance Number (c)** — Maximum number of defectives allowed in sample
- **Producer's Risk (α)** — Probability of rejecting a good lot (Type I error)
- **Consumer's Risk (β)** — Probability of accepting a bad lot (Type II error)
- **AQL** — Acceptable Quality Limit (producer's quality level)
- **LTPD** — Lot Tolerance Percent Defective (consumer's quality level)

Types of Acceptance Sampling Plans

- ① **Single Sampling Plan** — One sample is taken
- ② **Double Sampling Plan** — May take a second sample if first is inconclusive
- ③ **Multiple Sampling Plan**
- ④ **Sequential Sampling Plan** — Items inspected one by one until decision is made

Most Common: Single and Double Sampling Plans

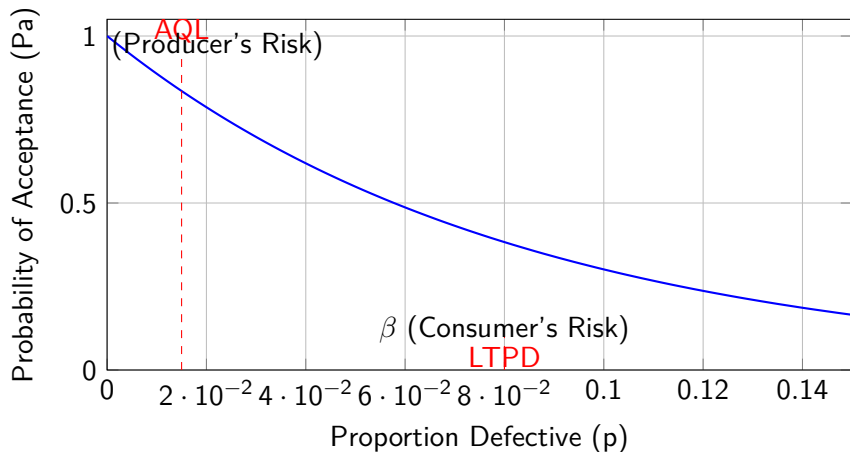
Operating Characteristic (OC) Curve

The OC Curve shows the **probability of accepting a lot** as a function of the actual quality level (proportion defective).

- Ideal OC Curve: Vertical line at AQL
- Real OC Curve: Shows discrimination power of the sampling plan

OC Curve – Visualization

Operating Characteristic (OC) Curve



Single Sampling Plan – Example

Parameters:

- Lot Size (N) = 2000
- Sample Size (n) = 125
- Acceptance Number (c) = 3
- AQL = 1.0%
- LTPD = 5.0%

Decision Rule:

- If number of defectives in sample $\leq 3 \rightarrow$ Accept the lot
- If number of defectives $> 3 \rightarrow$ Reject the lot

Advantages and Disadvantages of Acceptance Sampling

Advantages

- Less expensive than 100% inspection
- Less damage to products
- Useful for destructive testing
- Provides clear decision rules

Disadvantages

- Risk of accepting bad lots or rejecting good lots
- Does not improve the process
- Should not replace SPC
- Requires careful plan design

Six Sigma – Introduction

- Data-driven methodology aimed at reducing process variation
- Goal: Achieve **3.4 Defects Per Million Opportunities (DPMO)**
- Originated at Motorola (1986) by Bill Smith
- Popularized by General Electric under Jack Welch
- Combines statistical tools with business strategy

Six Sigma Performance Levels

Sigma Level	DPMO	Process Yield
6σ	3.4	99.99966%
5σ	233	99.977%
4σ	6,210	99.38%
3σ	66,807	93.32%
2σ	308,537	69.15%

DMAIC – The Six Sigma Roadmap

- ➊ **Define** — Define problem, goals, customer requirements (SIPOC, Voice of Customer)
- ➋ **Measure** — Collect baseline data, validate measurement system (Gauge R&R)
- ➌ **Analyze** — Identify root causes (Fishbone, Pareto, Hypothesis Testing, Regression)
- ➍ **Improve** — Develop and implement solutions (DOE, Kaizen, Pilot)
- ➎ **Control** — Sustain gains (Control Charts, Control Plans, Dashboards)

Six Sigma Roles & Certification

- **Champions** — Senior managers who sponsor projects
- **Master Black Belts** — Experts who train and mentor
- **Black Belts** — Full-time project leaders
- **Green Belts** — Part-time project team members
- **Yellow Belts** — Basic awareness

Lean Six Sigma

- Combines **Six Sigma** (variation reduction) with **Lean** (waste elimination)
- Focus on both quality and speed
- Key Lean Tools: 5S, Value Stream Mapping, Kanban, Poka-Yoke, Just-in-Time
- 8 Wastes (Muda): Transportation, Inventory, Motion, Waiting, Overproduction, Overprocessing, Defects, Unused Talent

Six Sigma in Big Data Analytics Context

- Improve quality of data pipelines and ETL processes
- Reduce model drift and prediction errors
- Optimize A/B testing and experimentation
- Improve reliability of real-time analytics systems
- Reduce Cost of Poor Data Quality
- Apply DMAIC to machine learning model performance

Key Takeaways

- Acceptance Sampling is for lot acceptance decisions with known risks
- OC Curve helps evaluate the performance of a sampling plan
- Six Sigma aims for near-perfect quality through variation reduction
- DMAIC is the structured problem-solving framework
- Lean Six Sigma combines speed and quality
- These tools are highly relevant for Big Data quality improvement projects

Total Quality Management (TQM) – Introduction

- A comprehensive management philosophy focused on continuous improvement of quality
- Involves **everyone** in the organization — from top management to frontline employees
- Shifts focus from product inspection to **process improvement** and **customer satisfaction**
- Foundation for modern frameworks like Six Sigma, Lean, and ISO 9001

Key Contributors to TQM

Expert	Major Contribution
W. Edwards Deming	14 Points for Management, PDCA Cycle, System of
Joseph Juran	Quality Trilogy (Planning, Control, Improvement), Pa
Philip Crosby	"Quality is Free", Zero Defects, Cost of Poor Quality
Kaoru Ishikawa	Fishbone Diagram, 7 QC Tools, Company-wide Qual
Genichi Taguchi	Robust Design, Loss Function

Core Principles of TQM

- **Customer Focus** — Internal & external customers
- **Total Employee Involvement** — Empowerment and teamwork
- **Process-Centered Approach** — Manage processes, not just outcomes
- **Integrated System** — All departments work toward common goals
- **Strategic & Systematic Approach**
- **Continual Improvement** (Kaizen)
- **Fact-Based Decision Making** — Use data and statistical tools
- **Effective Communication**
- **Leadership Commitment**

Deming's 14 Points for Management (Summary)

- Create constancy of purpose
- Adopt the new philosophy
- Cease dependence on inspection
- End awarding business on price tag alone
- Improve constantly the system
- Institute training on the job
- Institute leadership
- Drive out fear
- Break down barriers between departments
- Eliminate slogans and targets
- Eliminate numerical quotas
- Remove barriers to pride of workmanship
- Institute education and self-improvement
- Put everybody to work on the transformation

Cost of Poor Quality (COPQ)

Four Categories (Crosby & Juran)

- **Prevention Costs** — Training, process planning, quality audits
- **Appraisal Costs** — Inspection, testing, calibration
- **Internal Failure Costs** — Rework, scrap, downtime
- **External Failure Costs** — Warranty, returns, lost goodwill, reputation damage

Key Message: Prevention is much cheaper than failure.

TQM Implementation Framework

- 1 Top Management Commitment
- 2 Customer Focus & Voice of Customer (VoC)
- 3 Employee Training and Empowerment
- 4 Process Management & Standardization
- 5 Continuous Improvement (PDCA / Kaizen)
- 6 Fact-Based Management (SPC, Data Analytics)
- 7 Recognition and Reward Systems
- 8 Supplier Partnership

TQM in Big Data Analytics Context

- Data Quality Management as core part of TQM
- Quality of algorithms and machine learning models
- Real-time quality monitoring of data pipelines
- Continuous improvement of analytical processes
- Building quality culture in data science teams
- Integration with DevOps (DataOps / MLOps)
- Reducing "Data Debt" and technical debt

Benefits of TQM

- Higher customer satisfaction and loyalty
- Reduced costs and waste
- Improved process efficiency and productivity
- Better employee morale and engagement
- Enhanced competitive advantage
- Foundation for Industry 4.0 and data-driven organizations

Big Data Applications in SQC – Why It Matters

- Traditional SQC was designed for small samples and batch processes
- Modern industries generate massive volumes of high-velocity data (IoT, sensors, logs)
- Big Data enables **real-time**, **predictive**, and **automated** quality control
- Combines classical statistical tools with modern data engineering and machine learning
- Creates significant competitive advantage through reduced defects, lower costs, and higher customer satisfaction

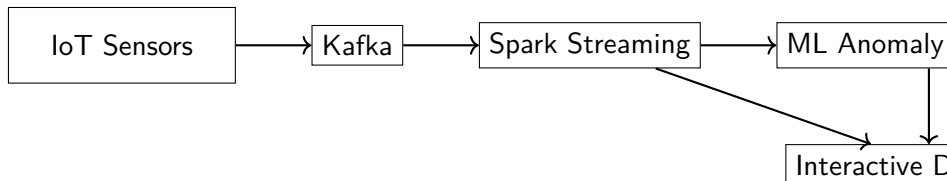
Traditional SQC vs Big Data-Enabled SQC

Traditional SQC	Big Data SQC
Small sample sizes	Millions of data points per minute
Periodic (batch) analysis	Real-time streaming analysis
Manual chart interpretation	Automated anomaly detection
Reactive	Predictive & Prescriptive
Limited to manufacturing	Applies to manufacturing, services, software, I

Real-Time Statistical Process Control

- Traditional control charts updated every few hours/days
- Big Data enables ****sub-second**** updates
- Technologies: Apache Kafka, Spark Streaming, Flink, Apache Storm
- Dynamic control limits that adapt to data volume and velocity
- Automated alerts and root cause analysis

Architecture for Real-Time SPC



Advanced Control Charts for Big Data

- **EWMA** (Exponentially Weighted Moving Average) — Sensitive to small shifts
- **CUSUM** (Cumulative Sum) — Excellent for detecting small persistent shifts
- Adaptive control limits using large historical data
- Multivariate Control Charts (Hotelling's T^2) for correlated variables
- Profile Monitoring for functional data (e.g., waveforms)

Machine Learning + SPC for Anomaly Detection

- Traditional 3-sigma rules + Modern ML algorithms
- Unsupervised techniques:
 - ▶ Isolation Forest
 - ▶ Autoencoders
 - ▶ One-Class SVM
- Hybrid approach: Use control charts for interpretability + ML for detection power
- Real-time scoring on streaming data

Large-Scale Process Capability Analysis

- Traditional: Small samples, normality assumption
- Big Data:
 - ▶ Non-parametric capability indices
 - ▶ Rolling capability analysis on sliding windows
 - ▶ Handling massive datasets with Spark / Dask
 - ▶ Capability analysis across multiple dimensions (multivariate)
- Predictive Capability using time-series forecasting

Predictive Quality Analytics

- Move from reactive to ****predictive**** quality
- Predict defects before they occur using:
 - ▶ Time-series forecasting
 - ▶ Survival analysis
 - ▶ Deep Learning on sensor patterns
- Digital Twin + Quality Prediction
- Remaining Useful Life (RUL) prediction for equipment

Case Study 1: Smart Manufacturing

Scenario: Automotive assembly line with 10,000+ sensors

- Real-time $\bar{X}$ and EWMA charts on Spark
- Anomaly detection using Isolation Forest
- Predictive maintenance reduced unplanned downtime by 35%
- Automated quality alerts sent to operators via mobile

Case Study 2: Data Pipeline Quality (Big Data ETL)

Problem: High error rate in daily ETL pipeline

- Implemented real-time p-chart on data validation results
- Used ML to detect anomalous data patterns
- Reduced error rate from 2.8% to 0.3%
- Built self-healing data quality rules

Challenges in Big Data SQC

- Volume, Velocity, Variety, Veracity
- High-dimensional data
- Non-stationary processes
- Data quality of the quality data itself
- Interpretability of black-box ML models
- Computational cost and scalability

Opportunities for Big Data Analysts

- Develop end-to-end quality monitoring platforms
- Build hybrid SPC + ML systems
- Create interactive quality dashboards (Power BI, Grafana, Superset)
- Design automated closed-loop quality control systems
- Research in areas like Deep SPC, Quantum SPC, and Edge Computing quality

Key Takeaways

- Big Data transforms SQC from periodic inspection to real-time intelligent monitoring
- Integration of classical tools (Control Charts, Capability) with modern technologies is powerful
- Predictive and prescriptive quality is the future
- Data quality itself must be managed as a core process
- MSc Big Data graduates are ideally positioned to lead quality transformation initiatives